1 - Intro to Chemistry

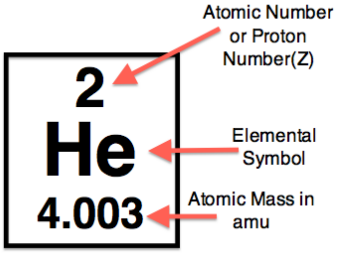
Matter is anything that has mass and volume found in 3 physical states (solid, liquid, gas)
Elements are the basic substances that make up all matter

Pure Substances	Mixtures
Elements: made up of only 1 type of atom, cannot be decomposed into simpler substances (e.g. Carbon, Oxygen) Compounds: 2+ different elements are chemically combined with fixed ratio (e.g. H_2O)	Heterogeneous/Mechanical Mixtures: all components are visible Homogeneous Mixtures/Solution: components are blended into a single substance

Physical Changes affect physical appearance of matter, but not its composition (e.g. ice melt)
Chemical Changes alter the composition of matter with the following properties:
New color appears, Heat/Light produced, Gas Bubbles produced, Precipitate forms in a liquid

Atoms are the smallest unit of a chemical element, the fundamental building block of matter. They are made up of even smaller (subatomic) particles, which are protons (p^+), neutrons (n^0), and electrons (e^-). Protons and neutrons cluster together to form the nucleus, while fast-moving electrons occupy the space that surrounds it.

Diatomic Molecules are two atoms of the same element: H_2 O_2 F_2 Br_2 I_2 N_2 Cl_2
Molecules are a group of atoms chemically bonded together by **covalent bonds** (e.g. covalent compounds), can be the same element or different elements

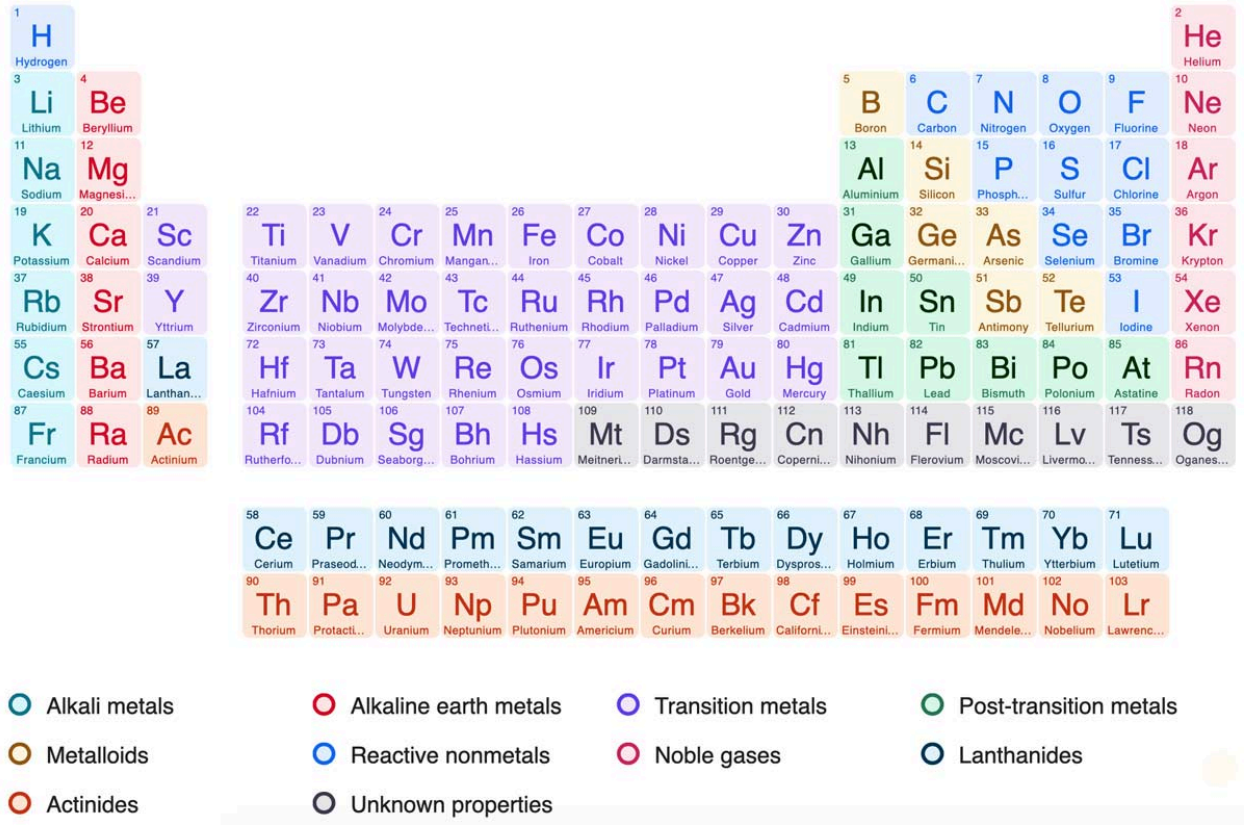


Atomic Number (Z) refers to the number of protons in an element
Mass Number (A) is the total number of protons and neutrons
Atomic Mass is the average mass of protons and neutrons
Average Atomic Mass is the weighted average of all isotopes, based on their abundance

Valence Electrons are electrons in the outermost energy level, responsible for the chemical behaviour of an element. Atom achieves stability with a full valence shell (Stable = Unreactive)

Charge means the imbalance of protons and electrons. It exists because atoms change their electron count to reach a more stable state.
Ion (Charged Particle) exists when a neutral atom (equal p^+ & e^-) gains or gives up electrons
Anion is an atom that gains electrons to be negatively charged
Cation is an atom that gives up electrons to be positively charged
*Metals form positive ions (cations) and non-metals form negative ions (anions)

2 - Periodic Table & Trends



Atomic Size (Radius)

a) Down a Group → Size Increases
Valence electrons occupy an energy level that is further and further from the nucleus. Thus, they experience less attraction from the nucleus

b) Across a Period → Size Decreases
Positive charge (Protons) on the nucleus increases across a period, with stronger force exerted to pull the outer electrons closer

Ionization Energy

Energy required to remove one electron from an atom

a) Down a Group → Required Energy Decreases
Valence electrons are farther from the positive force of the nucleus, making them easier to remove than the electrons in lower energy levels

b) Across a Period → Required Energy Increases
Attraction between nucleus and valence electrons increases, more energy is required to pull an electron away from its atom

Electron Affinity

Change in energy when an electron is added to an atom
(more stable with an extra electron = releases more energy)

a) Down a Group → Required Energy Decreases
   Elements at the top with small atoms have stronger force of attraction and can hold an extra electron easier (more stable) than the elements at the bottom
b) Across a Period (except Noble Gas) → Required Energy Increases
   Non-metals can become closer to stability with extra electrons (release more), while metals rather lose an electron than become farther from stability (release less)

Electronegativity

Ability to hold electrons tightly

a) Down a Group → E.N. Decreases
   Valence electrons are further from the nucleus attraction and harder to hold electrons
b) Across a Period→ E.N. Increases
   While energy level stays the same, protons increase with a stronger hold of electrons

Isotopes

Atoms of the same element but with different numbers of neutrons
Due to the imbalance ratio of protons to neutrons and nucleus being too heavy, not all isotopes are stable. To become more stable, the nucleus releases energy/particles in the form of radiation (alpha, beta, gamma). That's why it's called a radioisotope.

Avg. Atomic Mass = (mass of isotope #1)(% abund.) + (mass of isotope #2)(% abund.) + …

e.g. Potassium has 2 isotopes, K-39 and K-41. What are the % abundances?
Let x be the % abundance of K-39, i.e. (1 - x) = % abundance of K-41

$$39.098u = 39x + 41(1 - x)$$
$$39.098 = -2x + 41$$
$$1.902 = 2x$$
$$x = 0.951$$

∴ K-39 % abundance = 95.1%, K-41 % abundance = 4.9%

★ Examples

1. Arrange the following in increasing ionization energy
   (a) Be, Mg, Sr (b) Bi, Cs, Ba (c) Na, Al, S
2. Arrange the following in increasing electron affinity
   (a) O, S, Se (b) F, Cl, Br (c) N, O, F
3. Arrange the following in increasing electronegativity
   (a) Li, F, C (b) Na, Mg, Al (c) P, S, Cl
4. Arrange the following in increasing atomic radius
   (a) Na, Al, Cl (b) Li, Na, K (c) O, S, Se

1(a) Sr, Mg, Be (b) Cs, Ba, Bi (c) S, Al, Na 2(a) Se, S, O (b) Br, Cl, F (c) N, O, F 3(a) Li, C, F (b) Na, Mg, Al (c) P, S, Cl 4(a) Cl, Al, Na (b) Li, Na, K (c) O, S, Se

3 - Basic Nomenclature

Nomenclature is a set of rules to generate systematic names for chemical compounds

Ionic Compound	**Covalent/Molecular Compound**
- metal cation + non-metal anion or metal + polyatomic ion - bonded with ionic bond (electrons transfer) <u>Formula to Name</u> Cation + Anion with "ide" ending e.g. $MgCl_2$ → Magnesium Chloride <u>Name to Formula</u> Use Criss-Cross to determine the subscripts e.g. Calcium Fluoride → Ca^{+2} & F^{-1} → CaF_2	- non-metal + non=metal - bonded with covalent bond (sharing electrons) <u>Naming & Formula</u> Use prefix, Cation + Anion with "ide" ending e.g. SO_2 → Sulfur Dioxide Nitrogen Monoxide → NO <u>Prefixes</u> [1] Mono, [2] Di, [3] Tri, [4] Tetra, [5] Penta, [6] Hexa, [7] Hepta, [8] Octa, [9] Nona, [10] Deca

Peroxide
- compound that contains the peroxide linkage (O–O single bond)
- only happens when oxygen atoms are bonded with an **ionic bond**
- when oxygen is bonded to another oxygen, each oxygen "shares" the electrons equally
- e.g. H_2O_2 → Hydrogen Peroxide | Na_2O_2 → Sodium Peroxide | Li_2O_2 → Lithium Peroxide

*Rule: **Compounds must be electrically neutral** overall

Acids are compounds that release H^+ in water with aqueous (aq) as subscript

Binary Acids	**Oxyacids**
- hydrogen + non-metal $HCl_{(aq)}$ → Hydrochloric Acid $H_2S_{(aq)}$ → Hydrosulfuric Acid	- hydrogen + polyatomic ions (with oxygen) - -ate → -ic, -ite → -ous $H_2SO_{4\,(aq)}$ → Sulfuric Acid $H_2SO_{3\,(aq)}$ → Sulfurous Acid

Multivalent Cations are cations with more than one positive charge

Copper (Cu)	+, 2+	Manganese (Mn)	2+, 3+, 4+, 7+
Mercury (Hg)	+, 2+	Lead (Pb)	2+, 4+
Iron (Fe)	2+, 3+	Tin (Sn)	2+, 4+
Cobalt (Co)	2+, 3+	Chromium (Cr)	2+, 3+
Nickel (Ni)	2+, 3+	Titanium (Ti)	2+, 3+, 4+
Gold (Au)	+, 3+		

e.g. Tin (II) Oxide → SnO | Tin (IV) Oxide → Sn_2O_4 → SnO_2

e.g. $PbCl_4$ → Lead (IV) Chloride | CoS → Cobalt (II) Sulfide

Polyatomic Ions are nonmetals covalently bonded together, acting as one unit with a charge

NH_4^+	Ammonium ion	CO_3^{2-}	Carbonate ion
H_3O^+	Hydronium ion	SO_4^{2-}	Sulfate ion
IO_3^-	Iodate ion	SiO_3^{2-}	Silicate ion
BrO_3^-	Bromate ion	CrO_4^{2-}	Chromate ion
ClO_3^-	Chlorate ion	$S_2O_3^{2-}$	Thiosulfate ion
NO_3^-	Nitrate ion	$Cr_2O_7^{2-}$	Dichromate ion
$C_2H_3O_2^-$	Acetate ion	$C_2O_4^{2-}$	Oxalate ion
MnO_4^-	Permanganate ion	O_2^{2-}	Peroxide ion
CHO_2^-	Formate ion	PO_4^{3-}	Phosphate ion
OH^-	Hydroxide ion	BO_3^{3-}	Borate ion
CN^-	Cyanide ion	AsO_4^{3-}	Arsenate ion

Oxyanions of Halogens (Cl, Br, I) (N, A, P, S [only follow the –ate/–ite pairs])

O → Per-_ite O_2 → _ite O_3 → _ate O_4 → Hypo-_ate

e.g. Perchlorite (ClO^-), Chlorite (ClO_2^-), Chlorate (ClO_3^-), Hypochlorate (ClO_4^-)

Hydrate is an ionic compound with water molecules incorporated in its crystal lattice

$$Salt \cdot xH_2O$$

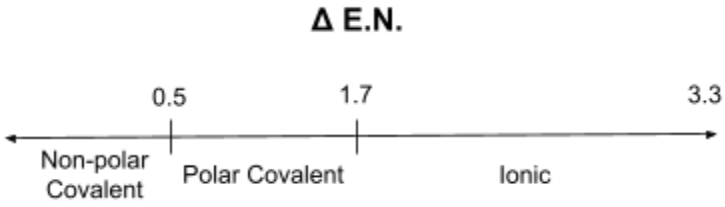
e.g. $CuSO_4 \cdot 5H_2O$ → copper(II) sulfate pentahydrate | $Na_2CO_3 \cdot 10H_2O$ → sodium carbonate decahydrate

★ Examples

*$Fe(NO_3)_3$ - Iron (III) Nitrate
PCl_3 - Phosphorus Trichloride
$Na_2Cr_2O_7$ - Sodium Dichromate
$HBr_{(aq)}$ - Hydrobromic Acid
N_2O_5 - Dinitrogen Pentoxide
$H_2SO_{3(aq)}$ - Sulfrous Acid
Cl_2O_7 - Dichlorine Heptaoxide
$8CoCl_2 \cdot 6H_2O$ - Cobalt Chloride Hexahydrate
*MnO_2 - Manganese (IV) Oxide
Na_2O_2 - Sodium Peroxide
$Cu(NO_2)_2$ - Copper (II) Nitrite
$FePO_4$ - Iron (III) Phosphate
$Pb(C_2H_3O_2)_4$ - Lead (IV) Acetate

Sulfur Hexafluoride - SF_6
Calcium Hydroxide - $Ca(OH)_2$
Phosphoric Acid - $H_3PO_{4(aq)}$
Tin (IV) Sulfate - $Sn(SO_4)_2$
*Chlorous Acid - $HClO_{2(aq)}$
Hydrocyanic Acid - $HCN_{(aq)}$
Dinitrogen Tetrahydride - $N_2H_{4(aq)}$
Potassium Permanganate - $KMnO_4$
Mercury (I) Chloride - Hg_2Cl_2
Carbon Tetrachloride - CCl_4
*Perchloric Acid - $HClO_4$
Ammonium Sulfate - $(NH_4)_2SO_4$
Cobalt (III) Hydroxide - $Co(OH)_3$

4 - Bondings & Forces



*Electronegativity (EN) = how strong an atom attracts bonding e^- (intramolecular bond polarity)

Intramolecular Bonds hold atoms together within molecules	**Intermolecular Forces** responsible for holding molecules together
<u>Ionic Bonds</u> An atom (non-metal) in the bond has greater force of attraction and takes electrons from the other (metal), creating ions of unlike charge held together by electrostatic force of attractions	<u>London Dispersion Forces (LDF)</u> Electrons move around to create temporary dipoles and weak forces that is present in all molecules (stronger in bigger ones)
<u>Covalent Bonds</u> Neither atom has strong enough E.N. to completely attract electrons, so the atoms are held together by the sharing of electrons **Equal sharing = Non-polar Covalent** **Unequal sharing = Polar Covalent**	<u>Dipole-Dipole Force (only in polar molecules)</u> Permanent dipole of one molecule attracts the opposite dipole of another. <u>Hydrogen Bonding</u> Special strong dipole-dipole force that happens when H is bonded with N, O, or F *Atoms share the hydrogen (proton), not electrons <u>Electrostatic Force (ion from ionic cpd. + polar)</u> Charged end of the polar molecule is drawn to the ion with the opposite charge (e.g. NaCl + H_2O)

Ionic Compound Properties	**Covalent Compound Properties**
- tend to exist in **crystal lattice structure** (highly organized 3D arrangement of atoms/molecules within a crystalline solid) - high melting/boiling points (strong bonds) - conduct electricity when liquid or aqueous - water soluble	- can be gases, liquids, or solids - moderate to low melting/boiling points - do not conduct electricity - some are soluble in water

<u>Indicating Polarity</u>

Atom with a stronger attraction for electrons (higher electronegativity) has a δ– charge ("slightly negative" end of the bond) while the other one has a δ^+ charge ("slightly positive" end of the bond).

Polarity describes how electrons are distributed in a bond/molecule

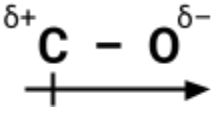
If one atom attracts electrons more strongly (higher electronegativity), the bond becomes polar.

Dipole is the separation of positive and negative charge in a bond or molecule. It is often shown as an arrow (→) pointing toward the δ– end, with a plus sign at the δ^+ end.

e.g. C - O

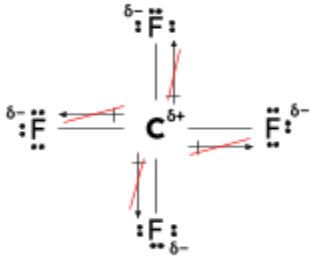
E.N.: C - 2.5

O - 3.5 ← stronger attraction (δ–)

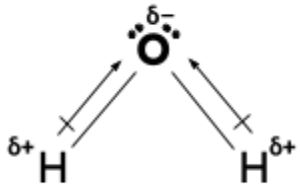


Molecule Polarity

Even if bonds are polar, the **shape of the molecule** decides if dipoles cancel or not
If dipoles are **symmetrical** → they cancel → molecule is nonpolar
If dipoles are **asymmetrical** → they add → molecule is polar



Vectors cancel = Non-polar Molecule



Polar Molecule

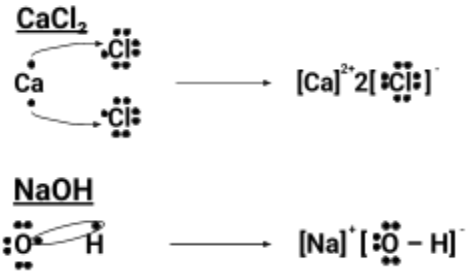
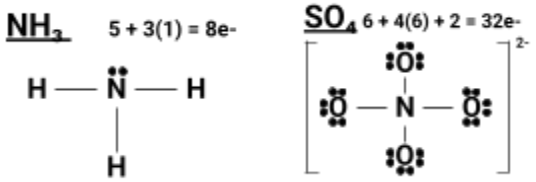
Non-polar Bond/Molecular: no partial positive/negative ends since electrons are equally shared
Polar Bond/Molecular: electrons are shared unequally causing dipoles (a "+" side and a "–" side)
*Molecule's overall polarity determines the intermolecular forces, which determines factors like solubility

Solubility

Solute = substance being dissolved (e.g.NaCl) | **Solvent** = substance doing dissolving (e.g. H_2O)
Solution = mixture of substances (e.g. Salt Water) *Rule: Solute dissolves in Solvent

Dissolving is a physical process where a substance separates into its component particles, with bonds broken and formed. If the solvent's interactions with solute are strong enough to overcome solute–solute and solvent–solvent forces, dissolving occurs.
> Polar molecules opposite charges are attracted and strong enough to break the existing intermolecular forces, solvent molecules then surround and interact with the solute
> In non-polar molecules, old dispersion forces break and new ones form between solute and solvent
*Rule: "Like dissolves like" - Polar mol. have too strong forces to overcome weak non-polar mol. forces

Lewis Structures is used to show the bonding of atoms in molecules

Ionic Compound	Covalent Compound
	Central Atom has the lowest electronegativity # of Bonds: H - 1 O - 2 N - 3 C - 4 

Ionic Compounds	Covalent Compounds	Acids
- Usually solids at room temp - If soluble in water, written as aqueous (aq) when dissolved - e.g. NaCl(s), $MgSO_4$(s)	Depends on molecule size + IMF - Small & Weak LDF: gases - Medium & Strong IMF: liquids - Large molecules: solids - e.g. $NH_{3(l)}$ (Hydrogen Bonding), $CO_{2(g)}$	- Written as aqueous when dissolved in water - Either gas/liquid without water - e.g. $HCl_{(g)}$ = hydrogen chloride gas $HCl_{(aq)}$ = hydrochloric acid

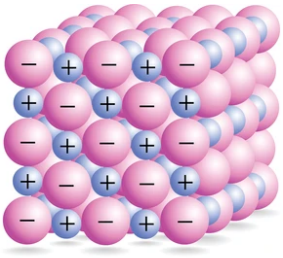
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Ionic Crystal

Positive ions and negative ions stick together, surrounding each ion and forming a 3D lattice

Properties
- Ionic Bonds
- High BP/MP
- Hard but brittle
- Liquid/Solution state high conductivity
- Polar solvent

e.g. NaCl, MgO



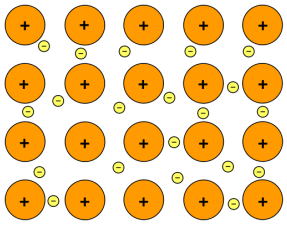
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Metallic Crystal

A lattice of positive metal ions surrounded by delocalized electrons that move freely

Properties
- Metallic Bonds
- Low-High BP/MP
- Shiny and flexible
- High conductivity in liquid/solution
- Non-polar solvent

e.g. Fe, Cu, Na, Al



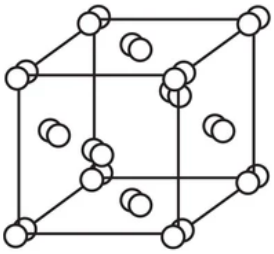
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Molecular Crystal

Made of individual molecules (not ions/ atoms) held together by intermolecular forces

Properties
- Intermolecular Force
- Low BP/MP
- Soft and do not conduct electricity
- Mostly nonpolar

e.g. CO_2, I_2, H_2O



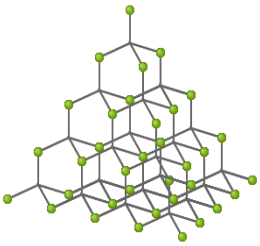
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Covalent Network Crystal

Nonmetal atoms joined by covalent bonds in a giant continuous network

Properties
- Covalent Bond
- High BP/MP
- Extremely hard
- Don't conduct elect.
- Insoluble

e.g. Diamond, Silicon

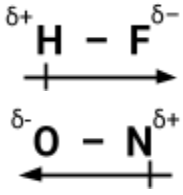


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★ Examples

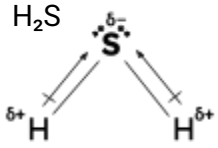
Determine the bond polarity of the following bonds (indicate dipoles [δ^+ and δ^-] if polar)

(a) C–H — Non-Polar
(b) O–O — Non-Polar
(c) H–F — Polar
(d) Cl–Br — Non-Polar
(e) O–N — Polar



Determine the polarity of the following molecules

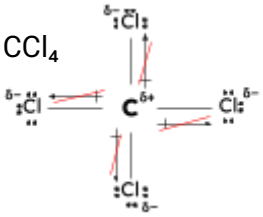
(a) H_2S



(b) CO_2



(c) CCl_4



Predict whether the following mixtures would dissolve in water or not

(a) NaCl — Dissolves (ion–dipole forces with H_2O)
(b) CH_4 — Insoluble (nonpolar, water is polar)
(c) $C_6H_{12}O_6$ (glucose) — Dissolves (–OH groups = hydrogen bonding with H_2O)
(d) I_2 — Insoluble (nonpolar molecule)

5 - Moles & Mass

Mole is a number to represent a large quantity of particles
Molar Mass is the mass of 1 mole of a substance measured in g/mol

1 mole = 6.022×10^{23} ← Avogadro's Constant (N_a)

Substance Formula	Mole Formula
$n = \frac{N}{N_A}$	$n = \frac{m}{M}$
n = moles (mol) N = # of particles N_A = Avogadro's constant	n = moles (mol) m = mass (g) M = molar mass (g/mol)

Percentage Composition
the percent, by mass, of each element in a compound

$$\frac{Molar\ mass\ of\ the\ element}{Molar\ mass\ of\ the\ compound} \times 100$$

★ Examples

1. How many moles do you have if you have 1.81×10^{24} atoms Cu?

   n = $\frac{N}{N_A}$ = $\frac{1.81 \times 10^{24}}{6.022 \times 10^{23}}$ = 3.01 mol

2. How many molecules and carbon atoms are in 40. Moles of sucrose ($C_{12}H_{22}O_{11}$)?

   n = $\frac{N}{N_A}$

   40. = $\frac{N}{6.022 \times 10^{23}}$

   N = 2.41×10^{25} molecules

   Number of carbon atoms:
   $= 2.41 \times 10^{25} \times 12$ C atom
   $= 2.9 \times 10^{26}$ atoms

3. How many moles of CF_4 are in a 1.55×10^5 kg sample?

   Molar Mass = 12.01 + 4(19.00) = 88.01 g/mol

   n = $\frac{m}{M}$ = $\frac{(1.55 \times 10^5) \times 1000}{88.01}$ = 1.76×10^6 mol

4. How many grams are in 5.26×10^{22} molecules of water?

   Molar Mass = 2(1.01) + 16.00 = 18.02 g/mol

   n = $\frac{N}{N_A}$ = $\frac{5.26 \times 10^{22}}{6.022 \times 10^{23}}$ = 0.08735 mol

   n = $\frac{m}{M}$ m = 0.08735×18.02 = 1.57 g

5. What is the percentage composition of H_3PO_4?

   Molar Mass = 3(1.01) + 30.97 + 4(16.00) = 98 g/mol

   %H = $\frac{3(1.01)}{98} \times 100 = 3.09\%$ %P = $\frac{30.97}{98} \times 100 = 31.6\%$

   %O = $100\% - 3.09\% - 31.6\% = 65.31\%$

6 - Stoichiometry

Empirical Formula shows the simplest whole-number ratio of atoms in a compound, while molecular formula shows the actual number of atoms of each element in a molecule

Determining Empirical Formula

Given %H = 3.09%, %P = 31.60%, %O = 65.31%

$$H : P : O$$
$$3.09g : 31.60g : 65.31g$$
$$\frac{3.09g}{1.01g/mol} : \frac{31.60g}{30.97g/mol} : \frac{65.31g}{16.00g/mol}$$
$$\frac{3.05941mol}{1.020342mol} : \frac{1.020342mol}{1.020342mol} : \frac{4.081875mol}{1.020342mol}$$
$$2.998 : 1 : 4.08$$

∴ Empirical Formula = H_3PO_4

Step 1: Assume total mass is 100g
Step 2: Divide each by the its molar mass
Step 3: Divide each dividend by the smallest
Step 4: Round or multiple the ratio to integer

Determining Molecular Formula

Given %C = 40.5%, %H = 4.60%, %O = 54.5%
Molar Mass is 176.14 g/mol

$$C : H : O$$
$$\frac{40.5}{12.01g/mol} : \frac{4.60g}{1.01g/mol} : \frac{54.5g}{16.00g/mol}$$
$$\frac{3.3721898mol}{3.3721898mol} : \frac{4.55446mol}{3.3721898mol} : \frac{3.40625mol}{3.3721898mol}$$
$$1 : 1.3506 : 1.01$$
$$3(1 : 1.3506 : 1.01)$$

∴ Emperical Formula = $C_3H_4O_3$

Empirical Mass = 3C + 4H + 3O = 88.07 g/mol

Multiplying factor = $\frac{molar\ mass}{empirical\ mass} = \frac{176.14}{88.07} = 2$

∴ Molecular Formula = $2(C_3H_4O_3) = C_6H_8O_6$

Stoichiometry

calculation of products and reactants in a chemical reaction

What mass of barium carbonate is produced from 2.0mol of sodium carbonate with barium chloride?

$$Na_2CO_3 + BaCl_2 \rightarrow 2NaCl + BaCO_{3\,(s)}$$ ← Make sure to balance the equation

$$\frac{BaCO_3}{Na_2CO_3} : \frac{1}{1} = \frac{x\ mol}{2.0\ mol}$$ ← Must be in moles

$$x = 2.0\ mol\ BaCO_3$$

Molar Mass = 37.33 + 12.01 + 3(16.00) = 197.34 g/mol

n = $\frac{m}{M}$ = 2.0×197.34 = 394.68g = 390g

∴ Mass of Barium Carbonate is 390g

Limiting Reagent

determine the amount of product that can form as the reactant runs out

Suppose 100.0g each of propane (C_3H_8) and oxygen is mixed and ignited. If complete combustion occurs, what mass of CO_2 should be produced?

$$C_3H_8 + 5O_2 \rightarrow 4H_2O + 3CO_2$$

$$Moles\ (O_2) = \frac{100.0g}{32.00g/mol} = 3.125mol \quad Moles\ (C_3H_8) = \frac{100.0g}{3(12.01) + 8(1.01)g/mol} = 2.2671mol$$

Step 1 - Finding Limiting Reagent

$$\frac{C_3H_8}{O_2} : \frac{1}{5} = \frac{2.2671\ mol}{x\ mol}$$ ← set one as unknown

$$x = 11.336\ mol\ O_2 \text{ required}$$

∵ There is not enough O_2 (3.125 mol < 11.336 mol)

∴ O_2 is the limiting reagent

Step 2 - Determining carbon dioxide produced

$$\frac{CO_2}{O_2} : \frac{3}{5} = \frac{x\ mol}{3.125\ mol}$$ ← Always use the limiting

$$x = 1.875\ mol\ CO_2 \text{ produced}$$

$$Moles\ (CO_2) = 1.875mol \times 44.01g/mol = 82.52g$$

∴ $82.52g\ of\ CO_2$ should be produced

7 - Concentration

Concentration is the quantity of solute dissolved in a solvent or solution (Symbol: [ ])

% Mass/Mass	% Volume/Volume	% Mass / Volume
Used when solid is dissolved in solid or solid in liquid	Used when liquid is dissolved in liquid	Used for medicine when solid solute in liquid solvent
$\%m/m = \frac{m_{solute}\,(g)}{m_{solution}\,(g)} \times 100$	$\%v/v = \frac{V_{solute}\,(mL)}{V_{solution}\,(mL)} \times 100$	$\%m/v = \frac{m_{solute}\,(g)}{V_{solution}\,(mL)} \times 100$

Parts per million (ppm) & Parts per billion (ppb)

used for very small concentrations (e.g. pollutants, impurities, trace chemicals)

$$ppm = \frac{m_{solute}\,(g)}{m_{solution}\,(g)} \times 10^6 \qquad ppb = \frac{m_{solute}\,(g)}{m_{solution}\,(g)} \times 10^9$$

Molar Concentration (Molarity) is how many moles of solute are dissolved per liter of solution

$$C = \frac{n}{V}$$

C = concentration (mol/L) n = moles of solute (mol) V = volume of solution (L)

e.g. If 0.212mol of iron (III) chloride is dissolved to make a 0.175L solution, find the molarity of the $[Cl^-]$.

$FeCl_{3\,(s)} \rightarrow Fe^{3+}_{(aq)} + 3Cl^-_{(aq)}$ $\quad [FeCl_3] = \frac{0.212mol}{0.175L} = 1.21\,mol/L$ $\quad [Cl^-] = 1.21 \times 3 = 3.63\,mol/L$

Net Ionic Equation is a simplified chemical equation that only shows the ions/molecules that are involved in a reaction, excluding "spectator ions" that do not change during the process

Case 1 - Sodium Nitrate + Potassium Chloride

Balanced Chemical Equation:

$NaNO_{3\,(aq)} + KCl_{(aq)} \rightarrow NaCl_{(aq)} + KNO_{3\,(aq)}$

Total Ionic Equation:

$Na^+_{(aq)} + NO_3^-{}_{(aq)} + K^+_{(aq)} + Cl^-_{(aq)} \rightarrow Na^+_{(aq)} + Cl^-_{(aq)} + K^+_{(aq)} + NO_3^-{}_{(aq)}$

Net Ionic Equation:

N/A (No Reaction since all ions are spectator ions)

Case 2 - Barium Chloride + Sodium Sulfate

Balanced Chemical Equation:

$BaCl_{2\,(aq)} + Na_2SO_{4\,(aq)} \rightarrow BaSO_{4\,(s)} + 2NaCl_{(aq)}$

Total Ionic Equation:

$Ba^{2+}_{(aq)} + 2Cl^-_{(aq)} + 2Na^+_{(aq)} + SO_4^{2-}{}_{(aq)} \rightarrow 2Na^+_{(aq)} + 2Cl^-_{(aq)} + BaSO_4^+{}_{(aq)}$

Net ionic Equation:

$Ba^{2+}_{(aq)} + SO_4^{2-}{}_{(aq)} \rightarrow BaSO_4^+{}_{(aq)}$

Spectator Ions: $Na^+_{(aq)} + Cl^-_{(aq)}$

Dilution is the process of decreasing the concentration of a solute by adding more solvent

Dilution Formula

$$C_1V_1 = C_2V_2$$

e.g. What volume is needed to produce 2.0L of 0.10mol/L sulfuric acid from 18mol/L stock solution?

$C_1 = 0.10mol/L$ $C_2 = 18mol/L$ $V_1 = 2.0L$ $V_2 = ?L$

$$C_1V_1 = C_2V_2$$
$$(0.10)(2.0) = (18)V_2$$
$$V_2 = 0.011$$

∴ 0.011L of the concentration should be used.

★ Examples

1. Health Canada's guideline for the maximum amount of Hg in a fish is 0.5ppm. A 1.6kg salmon was tested, it was found to contain 0.6mg of Hg. Is this salmon safe to eat?

   $ppm = \frac{m_{solute}(g)}{m_{solution}(g)} \times 10^6 = \frac{0.0006g}{1600g} \times 10^6 = 0.375$

   ∵ 0.375 < 0.5

   ∴ The salmon is safe to eat.

2. What is the molarity of $CuSO_4$ if 50.0g was dissolved in 250mL of water?

   Molar Mass: 63.55 + 32.06 + 4(16.00) = 159.62g/mol

   Moles: $\frac{50.0g}{159.62g/mol}$ = 0.3132mol

   $[CuSO_4] = \frac{0.3132mol}{0.25L} = 1.3mol/L$

   ∴ $[CuSO_4]$ is 1.3mol/L.

3. What is the maximum mass of precipitate expected when 325mL of 0.500 mol/L iron (III) chloride solution is mixed with 475mL of 0.400 mol/L sodium hydroxide solution?

   $FeCl_{3\,(s)} + 3NaOH_{(aq)} \rightarrow Fe(OH)_{3(s)} + 3NaCl_{(aq)}$

   $Moles\ (FeCl_{3\,(s)}) = 0.500mol/L \times 0.325L = 0.1625\ mol$

   $Moles\ (NaOH_{(aq)}) = 0.400mol/L \times 0.475L = 0.19\ mol$

   $\frac{FeCl_{3\,(s)}}{NaOH_{(aq)}} : \frac{1}{3} = \frac{0.1625mol}{x\ mol}$ $\frac{NaOH_{(aq)}}{Fe(OH)_{3(s)}} : \frac{3}{1} = \frac{0.19mol}{x\ mol}$

   $x = 0.4875\ mol\ NaOH_{(aq)}$ $x = 0.0633\ mol\ Fe(OH)_{3(s)}$

   ∴ $NaOH_{(aq)}$ is the limiting reagent

   Molar Mass $(Fe(OH)_{3(s)})$ = 55.85 + 3(16.00) + 3(1.01) = 106.88 g/mol

   Mass $(Fe(OH)_{3(s)}) = 0.06333\ mol \times 106.88\ g/mol = 6.77\ g$

   ∴ 6.77 g of precipitate is expected.

8 - Types of Reactions

Law of Conservation of Mass
mass of a closed system must remain constant over time and mass cannot be created or destroyed in a chemical reaction (total reactants mass = total products mass)

Synthesis Reactions
multiple substances combine to form compound

A + B → AB or A + BC → ABC

a. metal + non-metal → binary compound
$4Fe_{(s)} + 3O_{2(g)} \rightarrow 2Fe_2O_{3(s)}$

b. metal + water → metal hydroxide (base)
$Na_2O_{(s)} + H_2O_{(l)} \rightarrow 2NaOH_{(aq)}$

c. non-metal + water → oxyacid (acid)
$CO_{2(g)} + H_2O_{(l)} \rightarrow H_2CO_{3\,(aq)}$

Decomposition Reactions
compound breaks up into elements or simpler compound

AB → A + B or ABC → A + BC

a. binary compound → metal + non-metal
$2Fe_2O_{3(s)} \rightarrow 4Fe_{(s)} + 3O_{2(g)}$

b. oxyacid → non-metal + water
$H_2CO_{3\,(aq)} \rightarrow CO_{2(g)} + H_2O_{(l)}$

c. peroxide → oxygen + cation oxide
$2Na_2O_{2\,(s)} \rightarrow O_{2(g)} + 2Na_2O_{(s)}$

d. solid metal carbonates → CO_2 + metal oxide
$CaCO_{3\,(s)} \rightarrow CO_{2(g)} + CaO_2$

e. solid metal chlorates → O_2 + metal chloride
$2NaClO_{3\,(s)} \rightarrow 3O_{2(g)} + 2NaCl_{(s)}$

Activity Series

F	Li
Cl	K
Br	Na
I	Sr
S	Ba
	Ca
	Mg
	Al
	Mn
	Zn
	Cr
	Fe
	Co
	Ni
	Sn
	Pb
	H
	Sb
	As
	Cu
	Hg
	Ag

Single Displacement
one element replaces another in a compound
metal replaces cation, non-metal replaces anion
*use activity series to determine if a reaction proceeds

$2Al_{(s)} + Fe_2O_{3\,(aq)} \rightarrow 2Fe_{(s)} + Al_2O_{3\,(aq)}$
$Cl_{2\,(g)} + 2NaBr_{(aq)} \rightarrow 2NaCl_{(aq)} + Br_{2\,(l)}$
$Br_{2\,(s)} + CaCl_{2\,(aq)} \rightarrow NR$ (∵ chlorine is more reactive than bromine)

Double Displacement
reaction of 2 ionic compounds in water (aqueous)
where cation and anion switch places to form 2 new ionic compounds
only occurs if precipitate (s), gas (g), or covalent compound forms
*use solubility chart to determine states

$NaOH_{(aq)} + HCl_{(aq)} \rightarrow NaCl_{(aq)} + H_2O_{(l)}$
$CuSO_{4\,(aq)} + 2NaOH_{(aq)} \rightarrow Cu(OH)_{2\,(s)} + Na_2SO_{4\,(aq)}$

Extension

$H_2CO_{3\,(aq)} \rightarrow CO_{2\,(g)} + H_2O_{(l)}$ $\quad$ $H_2SO_{3\,(aq)} \rightarrow SO_{2\,(g)} + H_2O_{(l)}$
$H_2OH_{(aq)} \rightarrow NH_{3\,(g)} + H_2O_{(l)}$ $\quad$ $H_2SO_{4\,(aq)} \rightarrow SO_{3\,(g)} + H_2O_{(l)}$

$K_2CO_{3\,(aq)} + HNO_{3\,(aq)} \rightarrow H_2CO_{3\,(aq)} + KNO_{3\,(aq)} \rightarrow CO_{2\,(g)} + H_2O_{(l)} + KNO_{3\,(aq)}$

Combustion Reactions

occurs when a hydrocarbon reacts to oxygen gas

Complete Combustion	Incomplete Combustion
hydrocarbon fuel + $O_{2(g)} \rightarrow CO_{2(g)} + H_2O_{(g)}$	not enough oxygen gas to react with and other products (CO and C) are produced
$CH_{4\,(g)} + 2O_{2\,(g)} \rightarrow CO_{2(g)} + 2H_2O_{(g)}$ $C_6H_{12}O_{6\,(s)} + O_{2\,(g)} \rightarrow CO_{2(g)} + H_2O_{(g)}$	$4CH_{4\,(g)} + 7O_{2\,(g)} \rightarrow 2CO_{2(g)} + 2CO_{(g)} + 8H_2O_{(g)}$ $2CH_{4\,(g)} + 3O_{2\,(g)} \rightarrow CO_{2(g)} + C_{(s)} + 4H_2O_{(g)}$

Combustion Analysis

determine the empirical formula of a compound by measuring CO_2 and H_2O produced from burning a sample

$C_xH_y + O_{2\,(g)} \rightarrow CO_{2(g)} + H_2O_{(g)}$ $\qquad$ $C_xH_yO_z + O_{2\,(g)} \rightarrow CO_{2(g)} + H_2O_{(g)}$

e.g. A 0.2500g sample of compound containing carbon, hydrogen, and oxygen undergoes complete combustion to produce 0.3664g of CO_2 and 0.1500g of H_2O. Find the empirical formula.

Moles of C = $\frac{0.3664g\ CO_2}{44.01\ g/mol} \times \frac{1\ mol\ C}{1\ mol\ CO_2} = 0.00832538 mol\ C$ $\qquad$ Mass of C = $0.00832538 \times 12.01 = 0.0998878g$

Moles of H = $\frac{0.1500g\ H_2O}{18.02\ g/mol} \times \frac{2\ mol\ H}{1\ mol\ H_2O} = 0.0166482 mol\ H$ $\qquad$ Mass of H = $0.0166482 \times 1.01 = 0.01681468g$

Mass of O = 0.2500 - 0.0998878 - 0.01681468 = 0.1332g $\qquad$ Moles of O = $\frac{0.1332g}{16.00\ g/mol} = 0.008325$

$$C : H : O$$
$$\frac{0.00832538mol}{0.008325mol} : \frac{0.0166482mol}{0.008325mol} : \frac{0.008325mol}{0.008325mol}$$
$$1 : 1.998 : 1$$

∴ Empirical Formula = CH_2O

★ Examples

Determine the type of reaction, then predict the products and balance each reaction.

$C_3H_8(g) + 5O_2(g) \rightarrow 3CO_2(g) + 4H_2O(g)$	Combustion
$AgNO_3(aq) + NaCl(aq) \rightarrow AgCl(s) + NaNO_3(aq)$	Double Displacement
$CaCO_3(s) \rightarrow CaO(s) + CO_2(g)$	Decomposition
$C_2H_5OH(l) + 3O_2(g) \rightarrow 2CO_2(g) + 3H_2O(g)$	Combustion
$N_2(g) + 3H_2(g) \rightarrow 2NH_3(g)$	Synthesis
$CaCO_3(s) + 2HCl(aq) \rightarrow CaCl_2(aq) + H_2O(l) + CO_2(g)$	Double Displacement
$Cl_2(g) + 2KBr(aq) \rightarrow 2KCl(aq) + Br_2(l)$	Single Displacement
$Zn(s) + 2HCl(aq) \rightarrow ZnCl_2(aq) + H_2(g)$	Single Displacement
$2KClO_3(s) \rightarrow 2KCl(s) + 3O_2(g)$	Decomposition
$2Na(s) + Cl_2(g) \rightarrow 2NaCl(s)$	Synthesis
$2H_2O_2(aq) \rightarrow 2H_2O(l) + O_2(g)$	Decomposition

9 - Hydrates & Acids

Hydrate is an ionic compound that has water molecules trapped inside its crystal structure
Water is incorporated in a fixed ratio and can be removed by heating and leaving anhydrous salt

e.g. If you heat to dehydrate 50.0g of $CaCl_2 : 2H_2O$, what will be the mass of anhydrous salt?

$$\%CaCl_2 = \frac{CaCl_2}{CaCl_2 : 2H_2O} \times 100 = \frac{40.08 + 2(35.45)}{40.08 + 2(35.45) + 2(18.02)} \times 100 = 75.486\%$$

$$50.0g \times 75.486\% = 37.7\ CaCl_2$$

*make sure to include the water when finding molar masses

e.g. Given the following data, find the hydrated formula
Mass of empty beaker = 9.65g Beaker + $Ba(OH)_2 : XH_2O$ = 62.35g Beaker + $Ba(OH)_2$ = 38.27g

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Mass of $Ba(OH)_2$ = 38.27 - 9.65 = 28.62g Mass of H_2O = 62.35 - 38.27 = 24.08g

$Ba(OH)_2$	:	H_2O
$\frac{28.62g}{171.35g/mol}$	:	$\frac{24.08g}{18.02g/mol}$
$\frac{0.167026554mol}{0.167026554mol}$	:	$\frac{1.33629301mol}{0.167026554mol}$
1	:	8

∴ Formula: $Ba(OH)_2 : 8H_2O$

Acids are substances that produce H^+ ions (protons) when dissolved in water
Bases are substances that produce OH^- ions in water, or can accept H^+ ions
*H^+ ions produced in acids attach to H_2O to form H_3O^+ (hydronium ion)

Properties of Acid	Properties of Bases
- tastes sour & feels gritty - corrosive (react with metals, bases) - conduct electricity - pH <7	- tastes bitter & feels slippery/soapy - conduct electricity in water - neutralize acids to form salt + water - pH >7
Strong acids → fully break apart in water e.g. HCl, HNO_3, H_2SO_4 Weak acids → only partially ionize e.g. $HC_2H_3O_2$ (acetic acid in vinegar)	Strong bases → fully break apart in water. e.g. NaOH, KOH, $Ca(OH)_2$ Weak bases → partially dissociate in water. e.g. NH_3 (ammonia)

pH (potential of hydrogen) is a measure of the level of acidity/alkalinity of a solution
pOH (power of the hydroxide ion) is a measure of the level of basicity/alkalinity of a solution

$$pH = -log[H^+ \text{ or } H_3O^+]\ ([H^+] = 10^{-pH})$$

$$pOH = -log[OH^-]\ ([OH^-] = 10^{-pOH})$$



e.g. If the $[H^+] = 1 \times 10^{-3} mol/L$, determine pH.
pH = $-log[1 \times 10^{-3}] = 3$

e.g. Find the pH of a solution with pOH 0.
pH = 14 - pOH = 14 - 0.6021 = 13.40

10 - Gas Laws

Gases are greatly affected by changes in temperature, volume, and pressure
Kelvin (K) is used to calculate temperature for gases (℃ + 273.15 = K)
Pascal (Pa) is the standard unit of measurement for pressure
*standard atmospheric pressure: 1 atm = 101.3 kPa = 760 mmHg

Boyle's Law	Charles' Law	Gay-Lussac's Law
$P_1V_1 = P_2V_2$ $V \propto \frac{1}{P}$	$\frac{V_1}{T_1} = \frac{V_2}{T_2}$ $V \propto T$	$\frac{P_1}{T_1} = \frac{P_2}{T_2}$ $P \propto T$
Combined Gas Law $\frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2}$		

e.g. In a balloon of original volume 10.0L at 25.0℃, the pressure is 125kPa. If the temperature increases to 75℃ and the pressure decreases to 750.0kPa, what will be the volume of gas?

$P_1 = 125kPa$ $P_2 = 75.0kPa$
$V_1 = 10.0L$ $V_2 = xL$
$T_1 = 25.0℃ + 273.15 = 298.15K$
$T_2 = 75.0℃ + 273.15 = 348.15K$

$$\frac{(125kPa)(10.0L)}{(298.15K)} = \frac{(75.0kPa)x}{(348.15K)}$$
$$V_2 = 19.5L$$

Ideal Gas is any gas that conforms to the assumptions of Kinetic Molecular Theory and follows the gas laws exactly in order to avoid high pressure and low temperature

$$PV = nRT$$

Universal Gas Constant (R) = 8.314 $\frac{kPa \cdot L}{mol \cdot K}$ n = Moles

e.g. How many moles are contained in a 25.0L oxygen tank at 32.8℃ and 2.25atm?

$P = 2.25 \cdot 101.3 = 227.925kPa$
$V = 25.0L$
$n = x\,mol$
$R = 8.314\ \frac{kPa \cdot L}{mol \cdot K}$
$T = 32.8℃ = 305.95K$

$$PV = nRT$$
$$(227.925)(25) = x(8.314)(305.95)$$
$$n = 2.24mol$$

Dalton's Law of Partial Pressure states that the total pressure exerted by a mixture of non-reacting gases is equal to the sum of the partial pressures of the individual gases

$$P_{Total} = \sum P_{gas\,1} + P_{gas\,2} + \ ...$$

11 - Intro to Organic Chemistry

Organic Compounds are molecules with **carbon atoms** covalently bonded to other atoms. Carbon is special due to its small atomic radius and being able to form 4 chemical bonds
Hydrocarbon is an organic compound that only contains carbon and hydrogen atoms

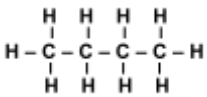
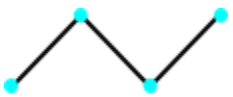
Type of Bond	Diagram	Name
Carbon – Oxygen	C – O	Ether
Carbon – Nitrogen	C – N	Amide
Carbon – Halogen	C – F C – Cl C – Br C – I	Organic Halide
Carbon – Hydroxyl Group	C – OH	Alcohol
Carbon – Amine Group	C – NH	Amine
Carbon – Oxygen (double bond)	C == O	Ketone, Aldehyde, Carboxylic Acid, Ester

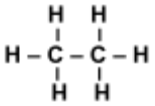
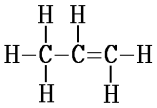
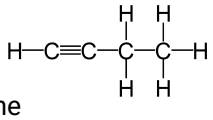
Nomenclatures

Prefix	# of Carbons
Meth	1
Eth	2
Prop	3
But	4
Pent	5
Hex	6
Hept	7
Oct	8
Non	9
Dec	10

Suffix	Carbon Bonds	Name
-ane	Single (C – C)	Alkane
-ene	Double (C == C)	Alkene
-yne	Triple (C ≡≡ C)	Alkyne

Prefix	# of Branches	Prefix	# of Branches
Mono	1	Penta	5
Di	2	Hexa	6
Tri	3	Hepta	7
Tetra	4	Octa	8

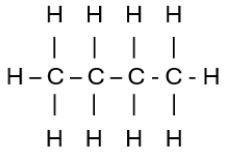
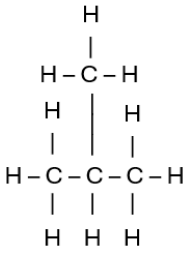
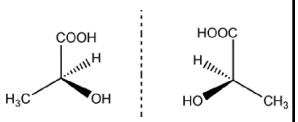
Expanded Structural Formula	Condensed Structural Formula	Line Structural Formula
	$CH_3 – CH_2 – CH_2 – CH_3$ or $H_3C – CH_2 – CH_2 – CH_3$	

Alkane	Alkene	Alkyne
Formula: C_nH_{2n+2} Name: Prefix + Suffix (prop + ane) e.g. n = 2 Formula - C_2H_6 Name - Ethane 	Formula: C_nH_{2n} Name: Prefix + # + Suffix (hex - 3 - ene) e.g. n = 3 Formula - C_3H_6 Name - Prop-1-ene 	Formula: C_nH_{2n-2} Name: Prefix + # + Suffix(but - 2 - yne) e.g. n = 4 Formula - C_4H_6  Name: But-1-yne

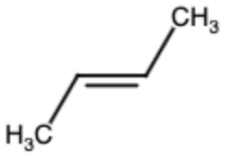
Alkyl Group is a functional group formed from an alkane by the removal of one hydrogen atom
Branching: alkanes can be straight chains or branched chains where alkyl groups are attached

Cyclic Hydrocarbon is a hydrocarbon with ring structures of alkanes, alkenes, and alkynes
Cycloalkanes General Formula: C_nH_{2n} Cycloalkanes Name: cyclo + prefix + suffix (e.g. cyclopentane)

Isomers are compounds that have the same chemical formula

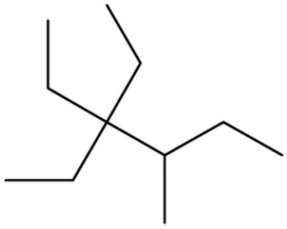
Structural Isomers	Stereoisomers
molecules that share the same molecular formula but different atom arrangements Different arrangement = Different connectivity = Different Chemical and Physical Properties C_4H_{10} $(C_4H_{2(4)+2})$  N-butane or butane  Isobutane or i-butane	Molecules with the same molecular formula and are bonded together in the same sequence **Enantiomers** mirror-image isomers that cannot be superimposed on each other because of chiral carbon (carbon with four different groups attached to it)  **Diastereomers** Stereoisomers that are not mirror images and not identical. It has at least one different stereocenter, resulting in different physical properties, such as melting point and boiling point.

Geometric Isomer is a type of diastereomer that occurs due to rotation blocked around a double bond (C=C). Each carbon in the double bond must have two different substituents.
If either carbon has two identical groups, no cis/trans possible.

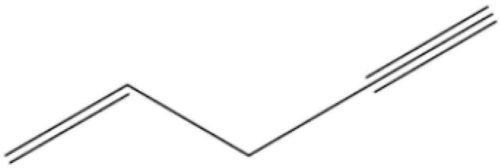
Cis Isomers	Trans Isomers
same side of the double bond  cis-but-2-ene	different side of the double bond  trans-but-2-ene

★ Examples

(a) 3,3-diethyl-4-methylhexane

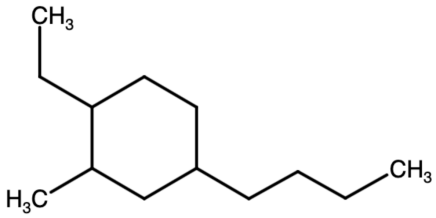


(b) Pent-1-en-4-yne

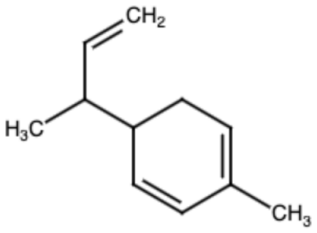


*double bond takes lower number

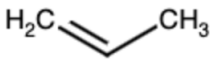
(c) 4-butyl-1-ethyl-2-methylcyclohexane



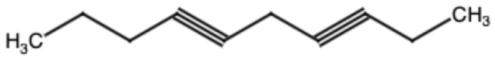
(d) 5-(but-3-en-2-yl)-2-methylcyclohexa-1,3-diene



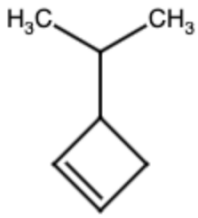
(e) propene



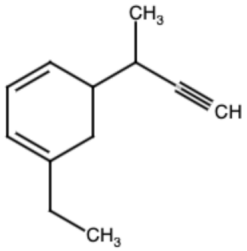
(f) dec-3,6-diyne



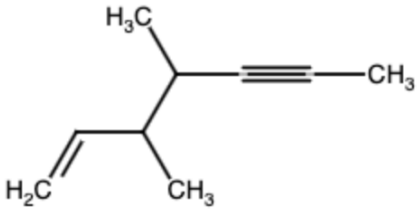
(g) 3-(propan-2-yl) cyclobut-1-ene



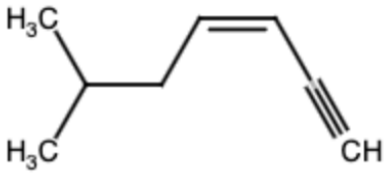
(h) 5-(but-3-yn-2-yl)-1-ethylcyclohexa-1,3-diene



(i) 3,4-dimethyl hept-1-en-5-yne

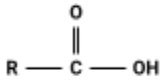
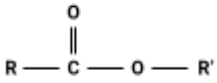
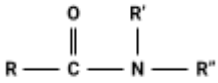
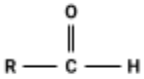
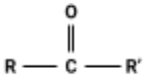
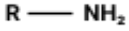
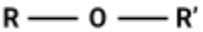
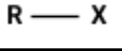


(j) cis-6-methylhept-3-en-1-yne



12 - Functional Groups

Functional Group is a particular atom combination that determines the substance's properties
Carbonyl Group (formyl group) is a function group with carbon double bonded with oxygen (C=O)

Priority Order	Structure	Name
Carboxylic Acid		-oic acid e.g. ethanoic acid (CH_3COOH)
Ester		-oate (alkoxycarbonyl) e.g. ethanoate (CH_3COO)
Amide		-amide e.g. ethanamide (CH_3CONH_2)
Aldehyde		-al (formyl/-oxo) e.g. ethanal (CH_3CHO)
Ketones		-#-one (-oxo) e.g. propan-2-one (CH_3COCH_3)
Alcohol		-#-ol (hydroxy) e.g. propan-1-ol ($CH_3CH_2CH_2OH$)
Amine		-#-amine (amino) e.g. propan-1-amine ($CN_3CH_2CH_2NH_2$)
Benzene		-benzene (phenyl) e.g. ethylbenzene ($C_6H_5CH_2CH_3$)
Alkene & Alkyne		-ene (C=C) / -yne (C≡C) e.g. ethene (C_2H_4) / ethyne (C_2H_2)
Ether		-oxy e.g. methoxymethane (CH_3OCH_3)
Halogen		iodo, bromo, chloro, fluoro
Alkane		-ane (C-C) (-yl) e.g. ethane (C_2H_6)

Benzene is a six-membered ring of carbon atoms with alternating double bonds
Phenyl is used to call a benzene ring when it is the substitute group

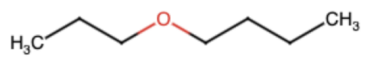
Aromatic Compounds	Aliphatic Compounds
- contain at least one benzene ring - obey Hückel's rule (conjugated, cyclic system) - e.g. benzene (C_6H_6), phenol (C_6H_5–OH)	- do not contain benzene rings - can be alkane, alkene, alkyne, or cyclic - e.g. propane (C_3H_8), cyclohexane (C_6H_{12})

Organic Halides are hydrocarbons with one or more hydrogen replaced by halogens (F, Cl, Br, I)
Numbering Order: Bromine (Bromo) → Chlorine (Chloro) → Fluorine (Fluoro) → Iodine (Iodo)

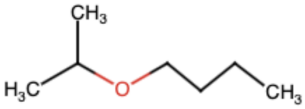
Ether (-oxy) is an organic compound with two alkyl groups attached to an oxygen atom

*add "oxy" to the **smaller** carbon chain

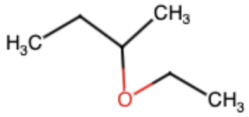
e.g. propoxy butane



e.g. 2-propoxy butane

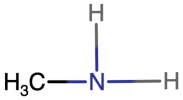
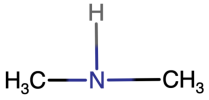
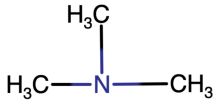


e.g. 2-ethoxy butane

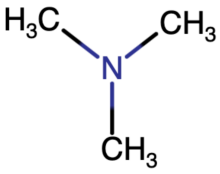


Amine (-amine) is an organic compound with ammonia (NH_3) attached to alkyl group(s)

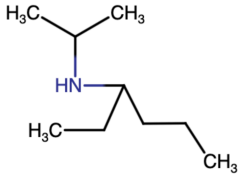
*add "amine" to the **longer** carbon chain

Primary (1°)	Secondary (2°)	Tertiary (3°)
		

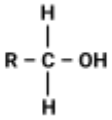
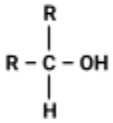
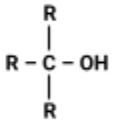
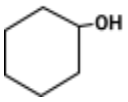
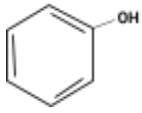
e.g. N,N-dimethyl methanamine



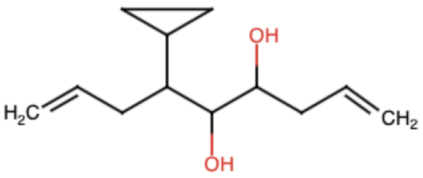
e.g. N-(propan-2-yl) hexan-3-amine



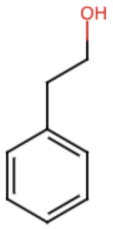
Alcohol (-ol) is an organic compound with the presence of a hydroxyl functional group (alcohol)

Primary (1°) OH attached C is attached to another C	Secondary (2°) OH attached C is attached to 2 other C	Tertiary (3°) OH attached C is attached to 3 other C	Cyclic alcohol that contains an alicyclic	Aromatic alcohol that contains a benzene as phenyl
				

e.g. cyclopropylnona-1,8-diene-4,5-diol

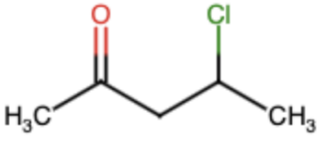


e.g. 2-phenyl ethan-1-ol

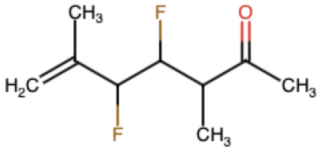


Ketone (-one) is an organic compound with carbonyl group bonded within the carbon chain

e.g. 4-chloropentan-2-one

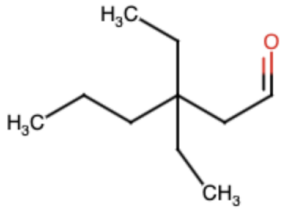


e.g. 4,5-difluoro-3,6-dimethylhept-6-en-2-one

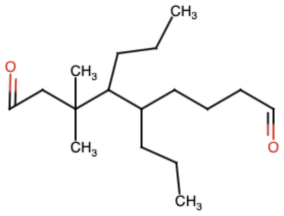


Aldehyde (-al) is an organic compound with carbonyl group bonded at the end of the chain

e.g. 4,4-diethyl hexanal

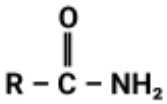
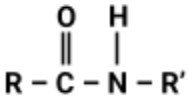
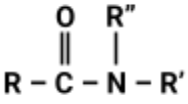


e.g. 3,3-dimethyl-4,5-dipropyl nonanedial

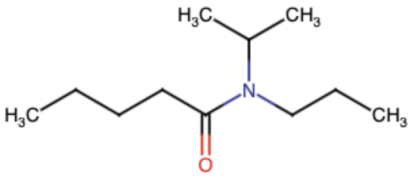


Amide (-amide) is an organic compound with carbon of carbonyl group bonded to nitrogen

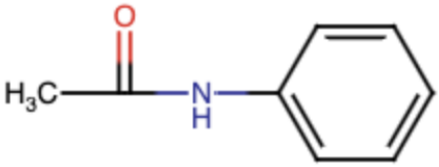
*add "amide" to the carbonyl group side

Primary (1°)	Secondary (2°)	Tertiary (3°)
		

e.g. N-(propan-2-yl)-N-propyl butanamide



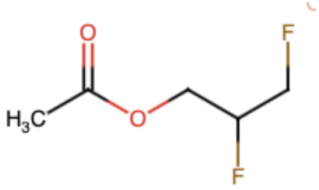
e.g. N-phenyl ethanamide



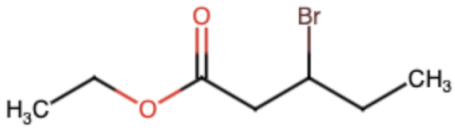
Ester (-oate) is an organic compound with a carbonyl group bonded to an oxygen

*add "oate" to the carbonyl group side

e.g. 2,3-difluoropropyl ethanoate

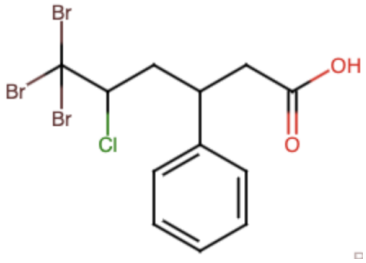


e.g. ethyl 3-bromopentanoate

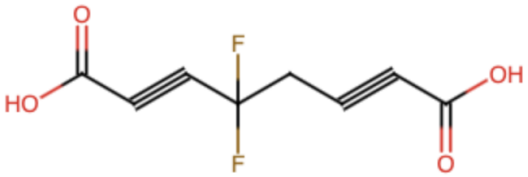


Carboxyl Group is a functional group that consists of hydroxyl group attached to carbonyl group
Carboxylic Acids are organic compounds characterized by the presence of a carboxyl group

e.g. 6,6,6-tribromo-5-chloro-4-phenylhexanoic acid



e.g. 4,4-difluoroocta-2,6-diynoic acid



Hydrogen Bonding	Dipole-Dipole Force	London Dispersion Force
Groups that have –OH or –NH bonds can directly H-bond - Carboxylic acids (–COOH) - Alcohols (–OH) - Amine & Amide ($–NH_2$/–NH) *only 1° or 2° amine/amide	Polar groups without –OH/–NH - Aldehydes (–CHO) - Ketones (–CO–) - Esters (–COOR) - Haloalkanes (–Cl, –Br, –I, –F)	Present in all molecules, but only force in nonpolar molecules - Alkanes (C–C / C–H) - Alkenes (C=C) - Alkynes (C≡C) - Aromatic rings (benzene)

Solubility - depends on how well molecules can interact with the solvent (H_2O)

Functional Group	Molecule Size
Hydrogen Bonded ⇒ Polar Soluble Dipole-Dipole (Carbonyl Group) ⇒ Slightly Polar Soluble Hydrocarbon (alkane/alkene/alkyne) ⇒ Hydrophobic	Small Alcohol (e.g. methanol) ⇒ Polar Soluble ↑ Hydrocarbon Chain ⇒ ↑ Non-Polar Character ⇒ Hydrophobic Effect ⇒ ↓ Polar Solubility

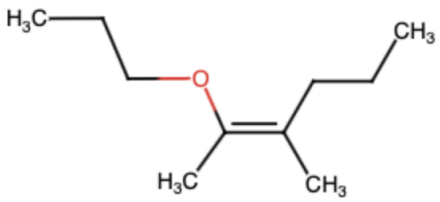
Molecular Packing
Straight Chains ⇒ larger hydrophobic region ⇒ ↓ Polar Solubility Branched Chains ⇒ smaller surface area ⇒ ↑ Polar Solubility

Melting/Boiling Points - intermolecular forces strength determines how much energy is needed

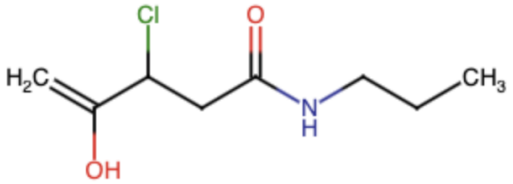
Functional Group	Molecule Size
Hydrogen Bonded ⇒ Stronger Bonds ⇒ High MP/BP Dipole-Dipole ⇒ Moderate MP/BP London Dispersion ⇒ Low MP/BP	Large Molecules ⇒ More Electrons ⇒ Stronger London Dispersion Force ⇒ Higher MP/BP
Molecular Packing	**Bonding**
Straight Chains ⇒ Pack Tightly ⇒ High MP/BP Branched ⇒ Reduce Area ⇒ Weak LDF ⇒ Low MP/BP	Double/Triple ⇒ Rigidity ⇒ Poor Pack ⇒ Low MP/BP Kinks (Cis) ⇒ Poor Pack ⇒ Low MP/BP

★ Examples

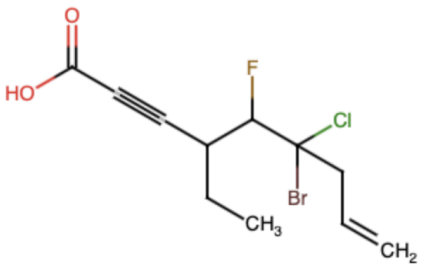
(a) cis-3-methyl-2-propoxyhex-2-ene



(b) 2-chloro-3-hydroxy-N-propylpent-3-enamide



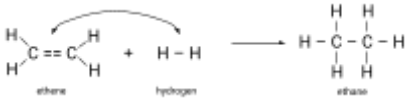
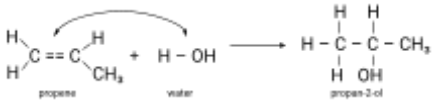
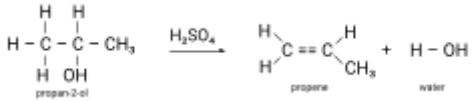
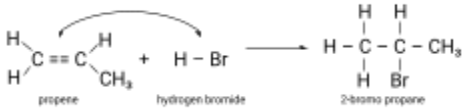
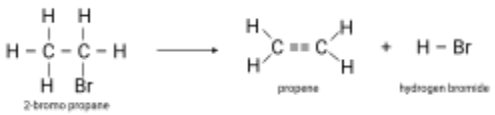
(c) 6-bromo-6-chloro-4-ethyl-5-fluoronon-8-en-2-ynoic acid

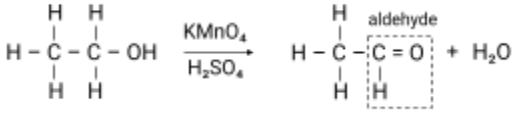
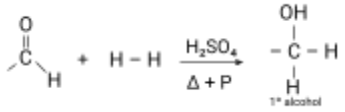
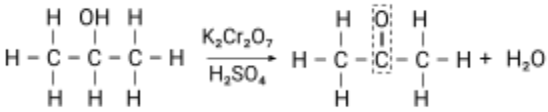
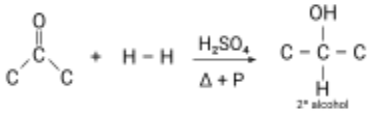
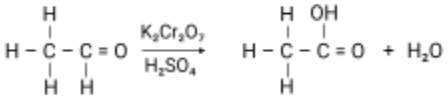
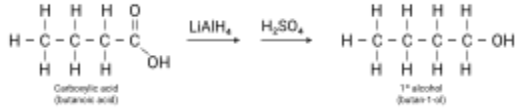
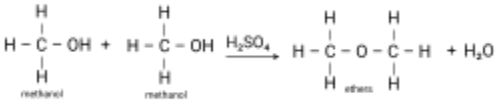
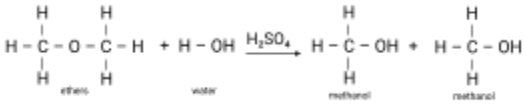


13 - Organic Reactions

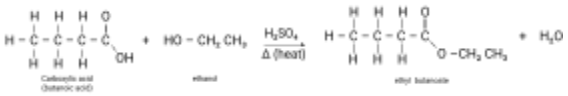
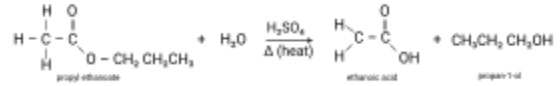
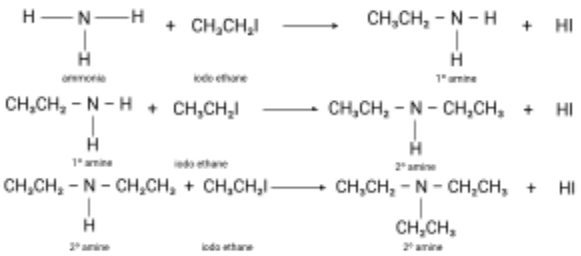
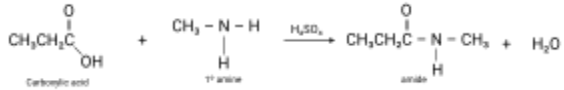
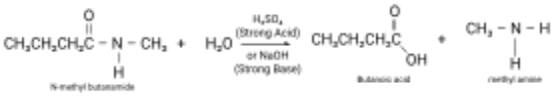
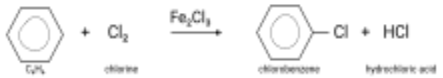
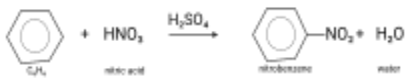
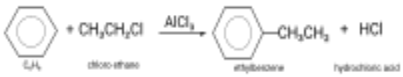
Saturated Fats have no double bonds in carbon chains with max possible H atoms (solid at RT)
Unsaturated Fats have one or more double bonds with fewer H atoms (liquid at RT)
Catalyst is a substance that speeds up a chemical reaction without being consumed
Markovnikov's Rule states that hydrogen from hydrohalogen or water bonds to the carbon within the double bond that has more hydrogens in an addition reaction (rich gets richer)

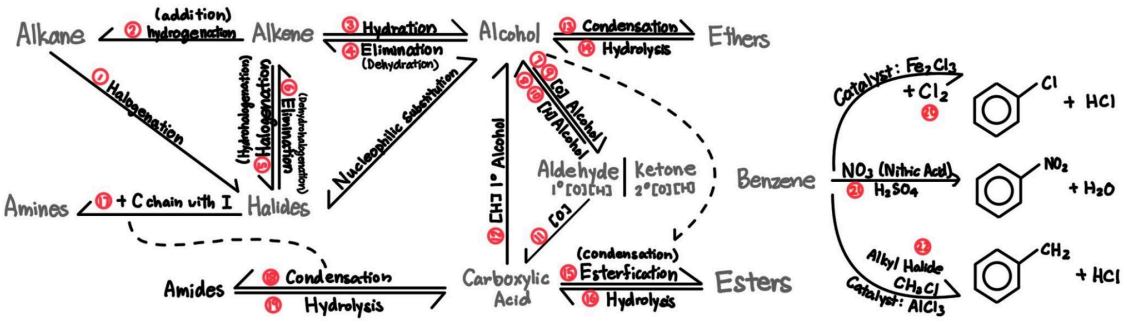
Redox Reaction stands for reduction and oxidation and they always come in pairs
Oxidation occurs when a carbon forms more bonds with oxygen and fewer with hydrogen
Oxidizing Agent [O]: $KMnO_4$ (Potassium Permanganate), $K_2Cr_2O_7$ (Potassium Dichromate), $Na_2Cr_2O_7$ (Sodium Dichromate)
Reduction occurs when a carbon forms less bonds with oxygen & more with hydrogen

#	Type	Reaction	Notes
1	Halogenation (Substitution) Alkane → Halides	Hydrogen atom is replaced by a halogen (Br, Cl, F)  	- Requires heat/UV - Radical chain reaction (keep reacting if there's Br and UV)
2	Hydrogenation (Addition) Alkene/yne → Alkane	+ Hydrogen  	- Requires a catalyst, heat, and pressure
3	Hydration (Addition) Alkene/yne → Alcohol	+ Water  	- Requires Catalyst (Conc. H_2SO_4) - Follows Markovnikov's Rule
4	Dehydration (Elimination) Alcohol → Alkene	Results in the removal of water  	- Occurs at 180ºC
5	Hydrohalogenation (Addition) Alkene/yne → Halide	+ HBr, HCl, HI  	- Requires room temperature - Follows Markovnikov's Rule
6	Dehydrohalogenation (Elimination) Halide → Alkene/yne	Removing hydrogen halide (H–X)  	- "Close down a bond"

7	Oxidation 1º Alcohol Alcohol → Aldehyde	H atom remains on C and produces aldehyde 	- Requires Catalyst (Conc. H_2SO_4) - [O]: $KMnO_4$
8	Reduction ⇒ 1º Alcohol Aldehyde → Alcohol		- Requires heat and pressure - Requires Catalyst (Conc. H_2SO_4)
9	Oxidation 2º Alcohol Alcohol → Ketone	carbonyl group formed is attached to 2 alkyl groups 	- Requires Catalyst (Conc. H_2SO_4) - [O]: $K_2Cr_2O_7$
10	Reduction ⇒ 2º Alcohol Ketone → Alcohol		- Requires heat and pressure - Requires Catalyst (Conc. H_2SO_4)
11	Complete Oxidation 1º Alcohol Aldehyde → Carboxylic Acid		- Requires Catalyst (Conc. H_2SO_4) - [O]: $K_2Cr_2O_7$ - Forms Carboxylic Acid from Aldehyde
12	Reduction ⇒ 1º Alcohol Carboxylic Acid → Alcohol		- Requires Catalyst (Conc. H_2SO_4) - [H]: $LiAlH_4$ / BH_3
13	Condensation Alcohol → Ethers	Joining reaction 	- Occurs at 140ºC - Requires Catalyst (Conc. H_2SO_4)
14	Hydrolysis Ethers → Alcohol	Uses water to cleave a molecule 	- Requires Catalyst (Conc. H_2SO_4)

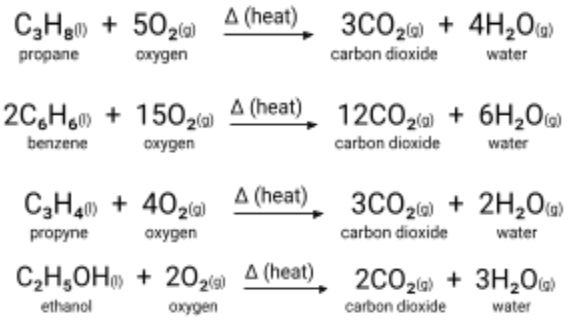
*Oxidation of 3º alcohol cannot proceed because it is bonded to 3 other carbons and has no H to form H_2O

15	Esterification (Condensation) Carboxylic Acid + Alcohol → Esters	Combine Carboxylic Acid and Alcohol to produce Ester and Water 	- Requires heat - Requires Catalyst (Conc. H_2SO_4)
16	Hydrolysis Esters → Carboxylic Acid	 *Reverse reactions are possible because conditions of esterification = conditions of ester hydrolysis	- Requires heat - Requires Catalyst (Conc. H_2SO_4)
17	Alkylation Halides → Amines		*Iodine is used due to its big size and readily donates alkyl chain *2º amine works too, but not 3º because no H available
18	Condensation Carboxylic Acid + Amines → Amides		- Requires Catalyst (Conc. H_2SO_4)
19	Hydrolysis Amides → Carboxylic Acid		- Requires Strong Acid (H_2SO_4) or Strong Base (NaOH)
20	Substitution Benzene + Cl_2		- Requires Catalyst (Conc. Fe_2Cl_3)
21	Substitution Benzene + Nitric Acid		- Requires Catalyst (Conc. H_2SO_4)
22	Substitution Benzene + Alkyl Halide		- Requires Catalyst (Conc. $AlCl_3$)



Combustion Reactions

1. Alkanes
   Alkanes are stable, but undergo combustion when ignited in air, making them useful fuels
2. Aromatics
3. Alkenes & Alkynes
   Alkenes and Alkynes are more reactive, and can undergo combustion just like alkanes
4. Alcohol



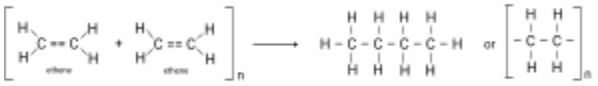
Polymer is a large molecule that consists of many repeating subunits (monomers).
Homopolymers is when all monomers are identical (-A-A-A-)
Heteropolymers consist of repeating pattern of different monomers (-A-B-A-B-A-)

Polymers are useful due to their characteristics, being resistant to chemicals, thermal & electrical insulators, and light in weight and strong structures.

- Polyethene (bottles & plastic bags): cheap, easy to process, flexible, strong
- Polyvinyl Chloride/PVC (construction & piping): change shape without breaking
- Polystyrene (protective packaging as trays & bottles): strong aromatic structure

Addition Polymerization

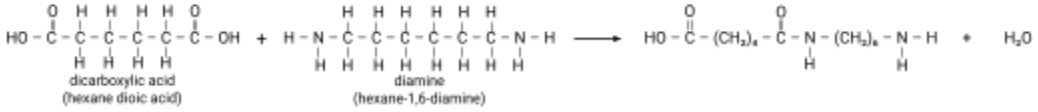
Unsaturated monomers are joined together and double bond breaks



Condensation Polymerization

Two different monomers are joined together through the elimination of H_2O

Polyesters → Carboxylic Acid + Alcohols Polyamides (Nylon) → Carboxylic Acids + Amines

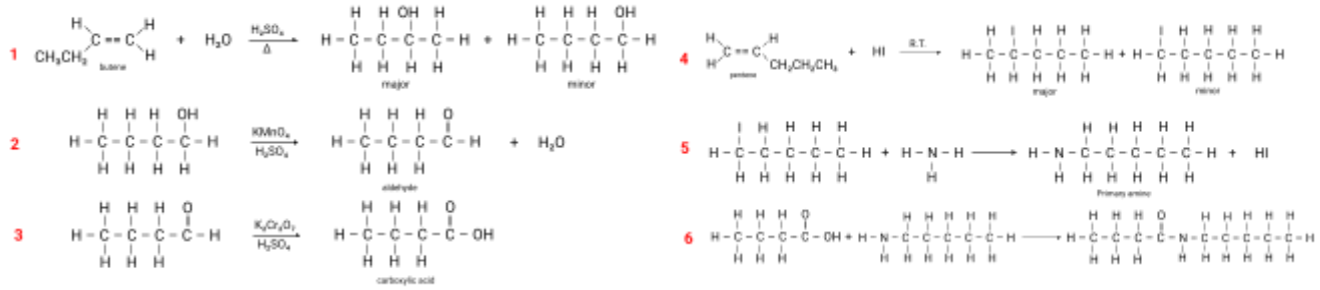


*Nylon 6,10 →

Polyamides with 6-carbon diamine + 10-carbon dicarboxylic acid

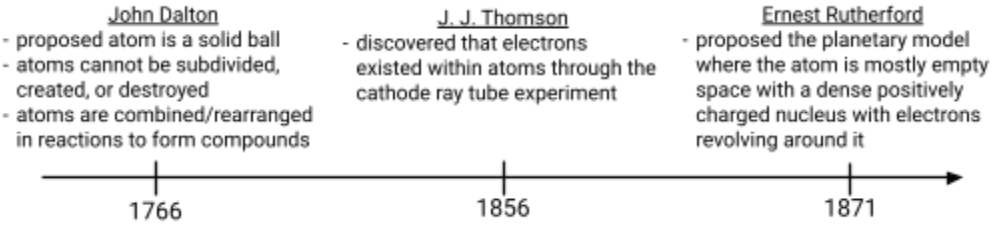
★ Examples

Propose a reaction pathway for the synthesis of N-pentylbutanamide starting with alkene



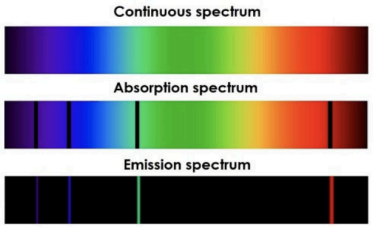
14 - Atomic Structure

History and Background

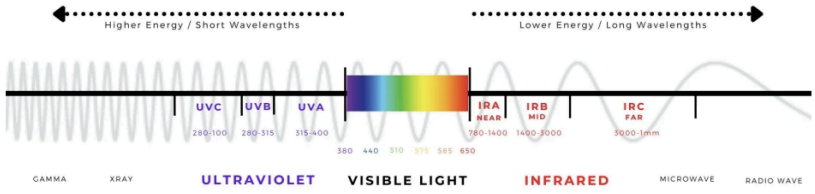


Quantum Theory

Quantum is a small discrete package of energy
Photon is a quantum of light energy that corresponds to a specific frequency and wavelength of light (E = hf)
*Shorter Wavelength (Red) corresponds to higher photon energy (λv = c)
*Higher Amplitude results in a brighter, more intense light ($I \propto A^2$)

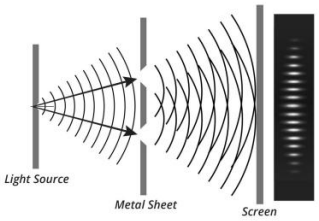


Photoelectric Effect - when light hits a metal surface, electrons are released only if the light's photons have enough energy. Below a certain frequency, no electrons are released.



With not enough energy to escape, electrons can absorb photons to move to higher shells (excited), and emit the same absorbed energy photons when fall back down (ground).

Light acts as a particle (photon) by carrying energy in discrete packets and transferring this energy in a process, shown by photoelectric effect. However, light can also behave like waves by creating interference patterns demonstrated in double slit experiment. This shows that light acts as both a wave and a particle — the **wave–particle duality**.



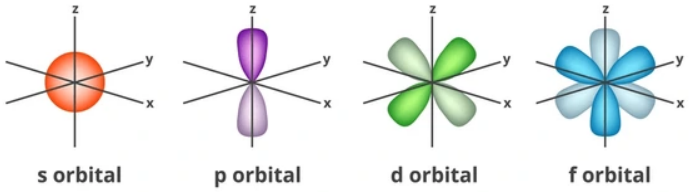
Heisenberg's Uncertainty Principle states that there is always uncertainty in either the position or momentum of a photon/electron. Increasing the precision of one will decrease the precision of the other.

Quantum Numbers

Orbitals are regions of highest probability in locating an electron with a given amount of energy
*each orbital (s, p, d, f) can only hold 2 electrons, one up-spin and one down-spin electron

Primary Quantum Number (n) represents the energy level (bigger number = high energy level = bigger atom)
Secondary Quantum Number (ℓ) specifies shape of region that electrons orbit around (s, p, d, f)
Magnetic Quantum Number (mℓ) describes the orientation/suborbital of the orbital (x, y, z)
Spin Quantum Number (m𝑠) describes a particle's spin created by magnetic fields
$m_s = +½$ ⇒ Clockwise (spin up) $m_s = -½$ ⇒ Anticlockwise (spin down)

Ferromagnetic Elements (Iron, Cobalt, Nickel) have unpaired electrons align into domains, which atoms are oriented with their poles in same direction to form permanent magnet



n	ℓ (0 to n-1)	mℓ (-ℓ to +ℓ)	Orbitals	Max Electrons
1	ℓ = 0	mℓ = 0	1 s-orbital	1 x 2 = 2e⁻ (when n=1)
2	ℓ = 0 ℓ = 1	mℓ = 0 mℓ = -1, 0, 1	1 s-orbital 3 p-orbital	(1+3) x 2 = 8e⁻ (when n=2)
3	ℓ = 0 ℓ = 1 ℓ = 2	mℓ = 0 mℓ = -1, 0, 1 mℓ = -2, -1, 0, 1, 2	1 s-orbital 3 p-orbital 5 d-orbital	(1+3+5) x 2 = 18e⁻ (when n=3)
4	ℓ = 0 ℓ = 1 ℓ = 2 ℓ = 3	mℓ = 0 mℓ = -1, 0, 1 mℓ = -2, -1, 0, 1, 2 mℓ = -3, -2, -1, 0, 1, 2, 3	1 s-orbital 3 p-orbital 5 d-orbital 7 f-orbital	(1+3+5+7) x 2 = 32e⁻ (when n=4)

Electron Configuration

C ⇒ $1s^2\ 2s^2\ 2p^2$

S ⇒ $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^4$

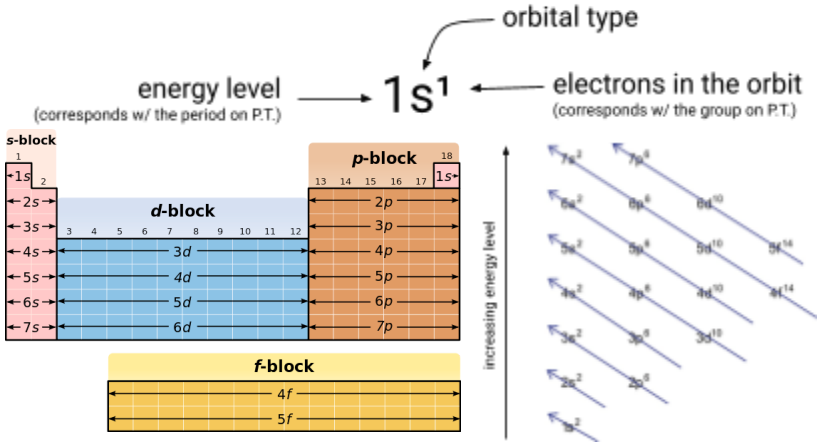
Co ⇒ $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^2\ 3d^7$

Ca ⇒ $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^2$

Ca^{2+} ⇒ $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6$

Cl ⇒ $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^5$

Cl^- ⇒ $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6$



Core Notation Electron Configuration

O ⇒ [He] $2s^2\ 2p^4$

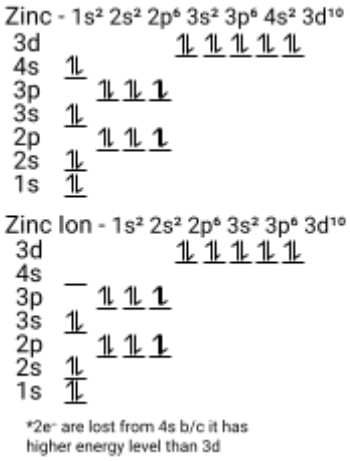
O^{2-} ⇒ [He] $2s^2\ 2p^6$

Sn ⇒ [Kr] $5s^2\ 4d^{10}\ 5p^2$

Sn^{2+} ⇒ [Kr] $5s^2\ 4d^{10}$

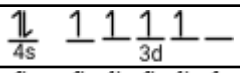
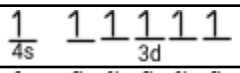
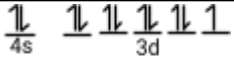
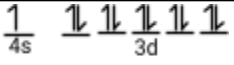
Energy Level Diagrams

Pauli Exclusion Principle	No two electrons in the same orbital can have the same spin (one spin-up, one spin-down)
Aufbau Principle	Electrons fill an atom in the lowest unoccupied atomic orbital first (P.T. order)
Hund's Rule	When multiple same energy orbitals are available, electrons spread out before being paired up



Electron Configuration Anomalies

Half-filled and fully filled subshells (d orbital) are more stable than unfilled subshells (s orbital)
This anomaly happens with Chromium (Cr), Molybdenum (Mo), Copper (Cu), Silver (Ag), Gold (Au)

	Prediction (Less Stable)		Actual (More Stable)	
Chromium (Cr)	[Ar] $4s^2\ 3d^4$		[Ar] $4s^1\ 3d^5$	
Copper (Cu)	[Ar] $4s^2\ 3d^9$		[Ar] $4s^1\ 3d^{10}$	

Lewis Structures

- Atoms/Ions are stable if they have a noble gas-like electron configuration (stable octet)
- Atoms form chemical bonds to achieve a stable octet
- Electrons are most stable when they are paired
- Exceptions: Be achieves stability with 4 electrons; B and Al achieve stability with 6 electrons
- Exceptions: Large elements starting from P3 (e.g. S, P, Cl) can hold more than 8 electrons

Resonance occurs when a molecule's bonding cannot be accurately represented by a single Lewis structure due to electron delocalization.
Formal Charge calculations are done to ensure electrons are balanced evenly throughout the molecule, finding the molecule most energetically stable. A formal charge closest to zero, with negative charges on more electronegative atoms, indicates the most stable Lewis structure.

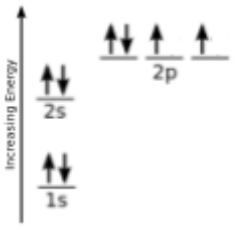
$$Formal\ Charge\ =\ (Valence\ electrons)\ -\ (Non\text{–}Bonding\ Valence\ Electrons)\ -\ (\frac{Bonding\ Electrons}{2})$$

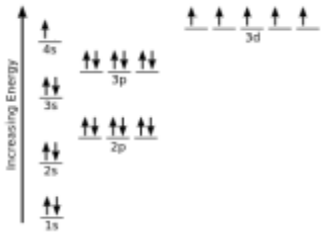
★ Examples

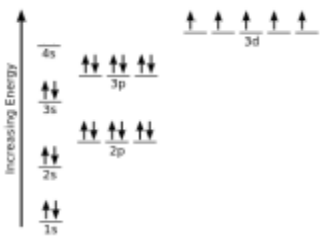
1. Write the full electron configuration for the following:
   a. Radium $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^2\ 3d^{10}\ 4p^6\ 5s^2\ 4d^{10}\ 5p^6\ 6s^2\ 4f^{14}\ 5d^{10}\ 6p^6\ 7s^2$
   b. Titanium $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^2\ 3d^2$
   c. Silver $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^2\ 3d^{10}\ 4p^6\ 5s^1\ 4d^{10}$
   d. Copper (II) Ion $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^9$

2. Write the core notation electron configuration for the following:
   a. Mercury [Xe] $6s^2\ 4f^{14}\ 5d^{10}$
   b. Arsenide [Ar] $4s^2\ 3d^{10}\ 4p^3$
   c. Bismuth [Xe] $6s^2\ 4f^{14}\ 5d^{10}\ 6p^3$
   d. Ununhexium [Rn] $7s^2\ 5f^{14}\ 6d^{10}\ 7p^4$

3. Draw the energy diagrams for the following:
   (a) Oxide (b) Chromium (c) Iron (III) ion





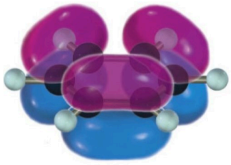


15 - Valence Theory

Valence Bond Theory

Atomic/Hybrid orbitals overlap to form a new orbital containing opposite spin electrons

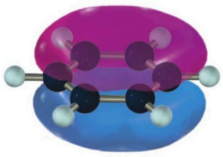
- Results in a decrease in energy ⇒ Stability ↑
- Greater overlap region ⇒ Stronger & stable bond
- Region of overlap has a maximum of 2 opposite spin electrons



Molecular Orbital Theory

New molecular orbitals are formed with new shapes and energy levels

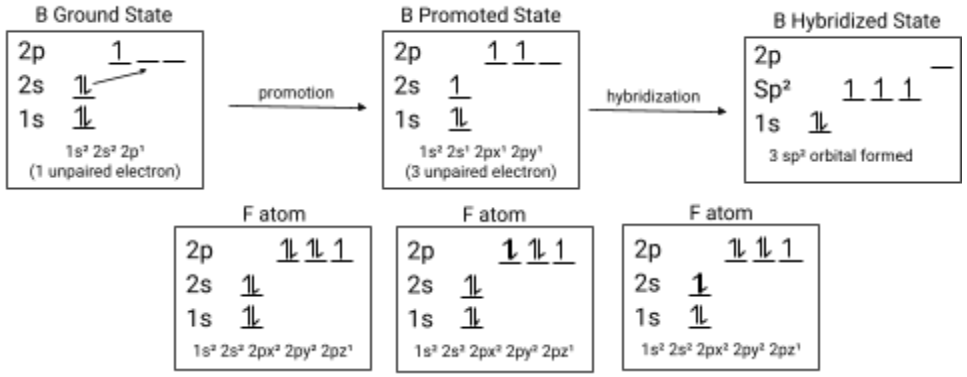
- Electrons are delocalized throughout the whole new orbital, rather than localized on a single atom
- Electrons fill the molecular orbitals in order of energy just like atomic orbitals



Hybridization is a theoretical process that mixes/hybridizes atomic orbitals to form new orbitals with correct shape and direction for bonding, explaining the formation of double & triple bonds

Hybrid Orbitals are obtained by combining different orbitals when bonding in the overlap region

BF_3



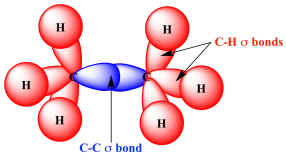
3 covalent bonds are formed by the overlap of 3 sp^2 hybrid orbitals from B atom with 2pz orbitals of 3 F atoms

$s^1 + p^1$ orbitals → sp hybrids | $s^1 + p^2$ orbitals → sp^2 hybrids | $s^1 + p^3$ orbitals → sp^3 hybrids

Sigma Bonds (σ)

Bond (single) created by the end-to-end overlap of atomic orbitals

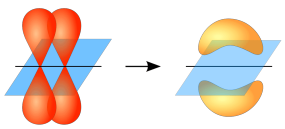
- formed first, only one per bond
- stronger than pi bond



Pi Bonds (π)

Bond (double/triple) created by the side-by-side overlap of atomic orbitals (p)

- results in 2 regions of electron density, above and below the common axis
- formed after sigma bonds, break easier



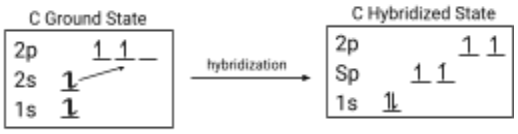
Double Bonds

formed by partial hybridization (unpaired electron in p orbital formed pi bond)
contain 1 sigma bond and 1 pi bond



Triple Bonds

formed by partial hybridization (unpaired electron in p orbital formed 2 pi bonds)
contain 1 sigma bond and 2 pi bonds



Total Domains	Bonded Atoms	Lone Pairs	Hybridization	Example
1	1	0	s	H_2
2	2	0	sp	CO_2
	1	1		CN^-
3	3	0	sp^2	$AlBr_3$
	2	1		$SnCl_2$
	1	2		O_2
4	4	0	sp^3	$SiCl_4$
	3	1		PH_3
	2	2		$SeBr_2$
	1	3		Cl_2
5	5	0	sp^3d	AsF_5
	4	1		SeH_4
	3	2		ICl_3
	2	3		BrF_2
6	6	0	sp^3d^2	$SeCl_6$
	5	1		IF_5
	4	2		XeF_4

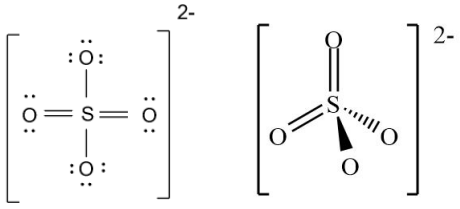
Valence Shell Electron Pair Repulsion (VSEPR) Theory predicts the shape of molecules with the valence electron pairs stay as far apart as possible to minimize repulsion of negative charges

A - Central atom | **X** - Bonded atom(s) attached to A | **E** - lone pair of electrons around A

Number of electron pairs	Electron pair geometries: 0 lone pair	1 lone pair	2 lone pairs	3 lone pairs	4 lone pairs
2	Linear				
3	Trigonal planar	Bent or angular			
4	Tetrahedral	Trigonal pyramid	Bent or angular		
5	Trigonal bipyramid	Sawhorse or seesaw	T-shape	Linear	
6	Octahedral	Square pyramid	Square planar	T-shape	Linear

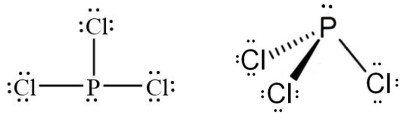
e.g. Predict shape of SO_4^{2-}

AX_4 = tetrahedral; sp^3



e.g. Predict shape of PCl_3

AX_3E = trigonal pyramidal; sp^3



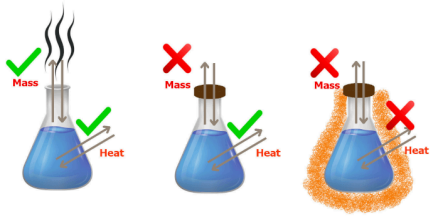
16 - Thermochemistry

Thermochemistry is the study of energy changes that accompany physical or chemical changes
Thermodynamics is the study of relationship between heat, work, energy, and temperature

1. Energy cannot be created or destroyed
2. If 2 systems are in equilibrium with third, they're in equilibrium with each other (A=B, B=C, A=C)

Calorimetry is the science of measuring the heat absorbed/released (during a chemical/physical change)

Open System	Closed System	Isolated System
Exchanges both energy and matter with its surrounding	Exchanges only energy, not matter (can't leave/enter)	Exchanges neither energy nor matter with its surroundings



Heat Transfer

Quantity of heat transferred (J) ⟵ $Q = mc\Delta T$ ⟵ Mass of substance (g), Change in temperature (°C) $\Delta T = Tf - Ti$

Specific Heat Capacity (c) - amount of heat required to raise a one gram substance by one degree (Water: 4.18 J/g °C)

Nuclear Reaction is a reaction that involves changes in the nuclei of atoms
Radioactive Decay is the process where an isotope with unstable atomic nucleus releases energy (in the form of particles or waves) to become more stable

Alpha Decay (α)	Beta Decay (ß)	Gamma Decay (γ)
- Loss of one α (He) particle - Lowest energy, can be stopped with paper	- Emits an electron (ß particle) - Medium energy, can be stopped by aluminum - Neutron → Proton	- Gamma photons emitted from high & excited nucleus - Highest energy, need blocks of lead/concrete to stop
$^{238}_{92}U \rightarrow {}^{234}_{90}Th + {}^{4}_{2}He$	$^{234}_{91}Pa \rightarrow {}^{234}_{92}U + {}^{0}_{-1}e$	$^{60}_{28}Ni \rightarrow {}^{60}_{28}Ni + \gamma$

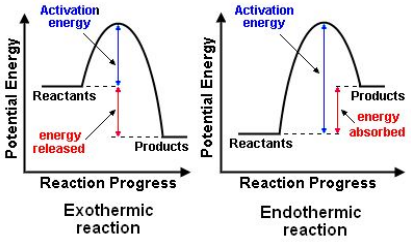
Nuclear Fission	Nuclear Fusion
process where a heavy atomic nucleus splits into two or more smaller nuclei, releasing a large amount of energy and neutrons	process where 2 light atomic nuclei combine to form a single, heavier nucleus, require high temperature, releases huge amount of energy
${}^{1}_{0}n + {}^{235}_{92}U \longrightarrow {}^{142}_{56}Ba + {}^{91}_{36}Kr + 3({}^{1}_{0}n)$	${}^{2}_{1}H + {}^{2}_{1}H \rightarrow {}^{3}_{2}He + {}^{1}_{0}n$

Enthalpy (H): measure of total energy of a thermodynamic system (measured in J or kJ)
Change in Enthalpy (ΔH): energy absorbed/released when system changes reactants→products

1. Physical Change (10^0 to 10^4 kJ) - Phase changes without a chemical reaction occurring
   e.g. Solute dissolves in solvent (ΔH_{sol}), Forming solid (ΔH_{fr}), Forming gas (ΔH_{vap})
2. Chemical Change (10^2 to 10^4 kJ) - Rearrangement of atoms and breaking bonds
   $\Delta H_{breaking\ bonds} = -\Delta H_{forming\ bonds}$
3. Nuclear Change (10^{10} to 10^{12} kJ) - change in protons and neutrons in an atom
   Mass of products is less as the missing mass is converted into energy

Endothermic (+ΔH): absorption of thermal energy as heat flows into the system (from surroundings)
e.g. Reactant + ΔH → Product

Exothermic (-ΔH): release of thermal energy as heat flows out of the system (into surroundings)
e.g. Reactant → Product + ΔH



Hess's Law: the enthalpy change of any reaction can be calculated with accumulated data
Molar Enthalpy (ΔH_x) is the enthalpy change (physical, chemical, nuclear) involving one mole

$$\Delta H_x\ (molar\ enthalpy) = \frac{\Delta H\ (change\ in\ enthalpy)}{n\ (mol)} \qquad \Delta H_{System} = -q_{surroundings}$$

Standard Molar Enthalpy of Formation ($\Delta H°_f$) is the change in enthalpy when one mole of compound is formed directly from its elements in their most stable state.

$$\Delta H = \Sigma n \Delta H°_{f\ (products)} - \Sigma n \Delta H°_{f\ (reactants)}$$

Bond Energies - all reactions require an activation energy

- Process of energy release: Break a bond (input energy) + Make new bond (release energy)
- Break weak bond, form strong bond (net release of energy)
- Break strong bond, form weak bond (net input of energy)

Overall Energy Change: ΔH_{rxn} = reactant bond energies (break) - products bond energies (form)

Rate of Reaction is the speed at which a chemical change occurs

$$r_{\text{average reaction rate}} = \frac{\Delta c\ \text{change in concentration}}{\Delta t\ \text{elapsed time}} \qquad (\text{Unit: } \frac{mol}{L \cdot S})$$

Rate of $A + 3B \rightarrow 2D$

$$\frac{\Delta[D]}{2\Delta t} = -\frac{\Delta[B]}{3\Delta t}$$

or

$$\frac{\Delta[D]}{\Delta t} = -\frac{\Delta[B]}{\Delta t} \times \frac{2mol\ D}{3mol\ B}$$

Instantaneous Rate of Reaction is speed at which a reaction is proceeding at a specific point of time (slope of tangent)

Negative sign = rate of consumption of reactant Positive sign = rate of production of product

<u>Methods of Measuring Reaction Rates</u>

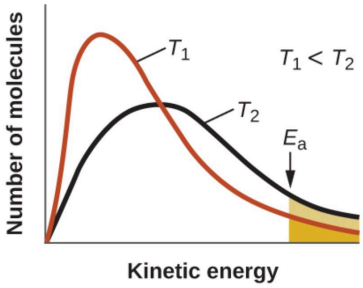
1. Color Change
   Measured by changes in light absorbency by spectrophotometer
2. Temperature Change
   Using calorimeter to simulate an isolated system and detect change in temperature
3. Pressure/Volume Change (for reactions that produce/consume gas)
   Detect pressure changes using barometer; Measure gas volume using collection tube
4. Mass Change
   Compare mass of reactants and products
5. pH Change (for acid/base reactions)
   Using pH meter
6. Ion Change
   Measuring the production of soluble ionic product due to conductivity using voltmeter

<u>Factors Affecting Rate of Reactions</u>

1. Temperature (more particles with minimum E_a)
   ↑temperature = ↑collisions =↑ rates
2. Surface Area (Solids) (e.g. solid to powder)
   ↑surface area =↑ expose particles =↑rate
3. Pressure (Gas)
   ↑pressure = ↓ volume = ↑concentration = more particles = ↑ rate
4. Catalyst
   Lowers activation energy with alternate pathway (∴ ↑ increased rate of reaction)
5. Reactants Concentration
   ↑particles = ↑overall collisions =↑rate
6. Reactants Chemical Nature
   Molecular compounds react slower than ionic compounds

Collision Theory explains that for a chemical reaction to occur, reactant particles must collide with sufficient energy (aka activation energy) and the correct orientation (rate ∝ # of collisions)
Activated Complex is an unstable chemical species containing partially broken and formed bonds, representing the maximum potential energy point in the change (top of graph)
Activation Energy is the minimum increase in potential energy of a system required for molecules to react (between reactant and activated complex)

Maxwell-Boltzmann Distribution is the distribution of kinetic energies (some move quickly and some slowly, but most are in mid-range of kinetic energy)
↑ temperature = more particles above the E_a (activation energy) = more particles to form products



Reaction Mechanism is a series of elementary steps that makes up an overall reaction
Elementary step is a step in a reaction mechanism involving one, two, or three particle colliding
Rate-determining step (RDS) is the slowest step in a reaction mechanism
Reaction Intermediates are molecules formed as short-lived products in reaction mechanism

Rate Law states that the rate is proportional to the products of the initial conc. of the reactants
Rate constant (k) is the proportionality constant in the rate law equation (temp. dependent)
Order of Reaction (n / m) is the exponent value that describes the initial conc. dependence of a particular reactant. Overall order of reaction is the sum of exponents in the rate law equation.

$$r = k[x]^m[y]^n \qquad k = Ae^{-Ea/RT}$$

Arrhenius Equation states that a small change in activation energy/temperature will have an exponential effect on the k value, and a large effect on the rate of reaction.

★ Examples

$$\frac{rate_1}{rate_2} = \left(\frac{conc_1}{conc_2}\right)^m$$

$2\ BrO_{3\,(aq)}^- + 5\ HSO_{3\,(aq)}^- \rightarrow Br_{2(l)} + 5\ SO_{4\,(aq)}^{2-} + H_2O_{(l)} + 3\ H_{(aq)}^+$

Trial	Initial $[BrO_3^-]$ (mol/L)	Initial $[HSO_3^-]$ (mol/L)	Initial Rate of Br_2 production (mol/L·s)
1	4.0×10^{-3}	6.0×10^{-3}	1.60×10^{-3}
2	2.0×10^{-3}	6.0×10^{-3}	0.80×10^{-3}
3	2.0×10^{-3}	3.0×10^{-3}	0.20×10^{-3}

BrO_3^-: $\frac{1.6\times10^{-3}}{6.8\times10^{-3}} = \left(\frac{4.0\times10^{-3}}{2.0\times10^{-3}}\right)^m$

$2 = 2^m$

$\log 2 = m \log 2$

$m = 1$

HSO_3^-: $\frac{0.8\times10^{-3}}{0.2\times10^{-3}} = \left(\frac{6.0\times10^{-3}}{3.0\times10^{-3}}\right)^n$

$4 = 2^n$

$\log 4 = n \log 2$

$n = 2$

$r = k[BrO_3^-]^1[HSO_3^-]^2$

$1.6 \times 10^{-3} = k(4.0 \times 10^{-3})^1(6.0 \times 10^{-3})^2$

$k = 1.11 \times 10^4 [BrO_3^-]^1[HSO_3^-]^2$

17 - Equilibrium Systems

Dynamic Equilibrium is a balance between forward and reverse processes occurring at the same rate, where concentrations are unchanging (achieved in a closed system)
Reversible Reaction is a reaction that can achieve equilibrium in the forward or reverse direction
*only temperature affects equilibrium

Solubility Equilibrium - dynamic equilibrium between solute and solvent in a saturated solution
Phase Equilibrium - dynamic equilibrium between different physical states of a pure substance
Chemical Reaction Equilibrium - dynamic equilibrium between reactants and products
Homogeneous Equilibrium - when all reactants and products are in the same physical state
Heterogeneous Equilibrium - when all reactants and products are in different physical states

Le Châtelier's Principle states that changes in concentration, temperature, volume or pressure will result in predictable shifts in the system in order to re-establish equilibrium.
Equilibrium Shift is the movement of a system at equilibrium, resulting in a change in the concentrations of reactants and products

Concentration	Temperature
Shifting in opposite direction for increase and same direction for decrease Increase reactant → Shift to the right Increase product → Shift to the left Decrease reactant → Shift to the left Decrease product → Shift to the right	Endothermic (heat in reactant, requires heat) Increase temperature → Shift to the right Decrease temperature → Shift to the left Exothermic (heat in product, produces heat) Increase temperature → Shift to the left Decrease temperature → Shift to the right
Volume/Pressure Increase volume → Decrease pressure → Shift in direction with more moles of gas Decrease volume → Increase pressure → Shift in direction with fewer moles of gas	

*Catalysts and addition of inert gases will result in no shift in equilibrium | Graphing Equilibrium Shifts

Equilibrium Law is a ratio of concentrations for the reversible reaction at equilibrium
Equilibrium Constant (K) is the calculated value of the concentration ratios at equilibrium
Reaction Quotient (Q) is the calculated value of the concentration ratios NOT at equilibrium
Solubility Product Constant (K_{sp}) describes the dissolution of a scarcely soluble ionic compound, equal to the product of the ion concentrations in a saturated solution
Recall: Solubility is the concentration of a saturated solution of solute in particular solvent

$$aA + bB \rightleftharpoons cC + dD \qquad M_xN_y \rightleftharpoons xM^{y+} + yN^{x-}$$

$$k = \frac{[C]^c[D]^d}{[A]^a[B]^b} \qquad Q = \frac{[C]^c[D]^d}{[A]^a[B]^b} \qquad K_{sp} = [M^+]^x + [N^-]^y$$

*k provide no information about the rate; only the equilibrium position of the reaction; only change when temperature changes
*value of any equilibrium constant (K, K_{sp}, K_w, K_a, K_b)depends on temperature
K = 1 indicates [reactants] = [products], K > 1 indicate products favoured, K < 1 indicate reactants favoured
K_{sp} << 1 indicates low solubility (reactants favoured), K_{sp} >> 1 indicate high solubility (products favoured)

Q = K → System is at equilibrium
Q < K → System is before equilibrium (shifts right)
Q > K → System is after equilibrium (shifts left)

Q = K_{sp} → Saturated solution (no precipitate)
Q < K_{sp} → Unsaturated solution (no precipitate)
Q > K_{sp} → Supersaturated solution (precipitate)

Common Ion Effect is a reduction in solubility of an ionic compound caused by the presence of another ionic compound with a common ion.

Arrhenius Theory: Base releases hydroxide ions while acid releases hydrogen ions in solution
Bronsted-Lowry Theory: Base accepts a hydrogen ion (proton) and acid donates a hydrogen ion, producing conjugate acid-base pairs (two molecules linked by the transfer of a proton)
Lewis Theory of Acids and Bases: Base donates electron pair while acid accepts electron pair

Amphoteric is a generic term for substances that can react as both acid and base (e.g. ZnO, Al2O3)
Amphiprotic is a specific subset of amphoteric substances that specifically donate/accept protons, following the Bronsted-Lowry Theory (e.g. H_2O, HCO_3^-, $H_2PO_4^-$)

Conjugate Acid-Base Pair are two substances whose formulas differ by only one H^+

Concentrated vs Dilute	Strong vs Base
describe the amount of acid/base dissolved in water (how much is present)	describe how much acid/base dissociates in water (how completely it reacts)
Concentrated: lots of solute in solution	Strong: Dissociates 100% (form H^+ ions)
Dilute: little amount of solute in solution	Weak: Dissociates partially (remain intact)

*analogy: Strong = how loudly each person shouts, Concentrated = how many people are in the room

Strong Acid: Acid that completely dissociates in water to form ions (good conductor of electricity)
Weak Acid: Acid that partially dissociates in water to form ions (poor conduct of electricity)
e.g. Strong Acid: HCl (Hydrochloride Acid), HBr (Hydrobromic Acid), HI (Hydroiodic Acid), HNO_3 (Nitric Acid), $HClO_3$ (Chloric Acid), $HClO_4$ (Perchloric Acid), H_2SO_4 (Sulfuric Acid)
Strong Base: Base that completely dissociates in water to form ions (good conductor of electricity)
Weak Base: Base that partially dissociates in water to form ions (poor conduct of electricity)
e.g. Strong Base: Group IA and Group IIA hydroxides; e.g. NaOH (Sodium Hydroxide), $Mg(OH)_2$ (Magnesium Hydroxide)

Monoprotic Acid: One H^+ (HCl) Diprotic Acid: Two H^+ (H_2SO_4)
Triprotic Acid: Three H^+ (H_3PO_4) Polyprotic Acid: Many H^+

Autoionization of Water is the self-ionization process where H_2O acts as both acid and base
Ion Product Constant (K_w) is an equilibrium constant for ionization of water (1.0×10^{-14})
Acid Ionization Constant (K_a) is an equilibrium constant for ionization of a weak acid
Base Ionization Constant (K_b) is an equilibrium constant for ionization of a weak base

$$K_w = K_a \times K_b = [H_3O^+]\ [OH^-] = 1.0 \times 10^{-14}$$

$$HA_{(aq)} + H_2O_{(l)} \rightleftharpoons A^-_{(aq)} + H_3O^+_{(aq)} \qquad K_a = \frac{[A^-_{(aq)}][H_3O^+_{(aq)}]}{[HA_{(aq)}]}$$

$$A^-_{(aq)} + H_2O_{(l)} \rightleftharpoons HA^+_{(aq)} + OH^-_{(aq)} \qquad K_b = \frac{[HA^+_{(aq)}][OH^-_{(aq)}]}{[A^-_{(aq)}]}$$

Percent Ionization: $\frac{Concentration\ of\ Conjugate\ Acid/Base}{Concentration\ of\ Weak\ Acid/Base} \times 100\%$

Recall: pH = -log[H_3O^+] pOH = -log[OH^-] pH + pOH = 14 Acidic: [H_3O^+] > [OH^-] Basic: [H_3O^+] < [OH^-] Neutral: [H_3O^+] = [OH^-]
*Strong acids have large K_a and Strong bases have large K_b
*conjugate base of strong acid → very weak base
conjugate base of weak acid → weak base
conjugate base of very weak acid → strong base
conjugate acid of strong base → very weak acid
conjugate acid of weak base → weak acid
conjugate acid of very weak base → strong acid

Hydrolysis is a chemical process where a compound is broken down by reacting with water, resulting in an acidic/basic solution (Soluble salts can undergo hydrolysis and change the solution's pH)
*Salts can be neutral (do not reacts with water, pH = 7), acidic (react with water, pH < 7), or basic (react with water, pH > 7)
Recall: Spectator Ions don't interact with water (aka cation from strong bases and anion from strong acids)

Salts (form acidic sol.)	Salts (form basic sol.)
- Cation from weak bases and anion from strong acids (e.g. NH_4Cl, N_2H_5Br) - Highly charged small cations, produce H^+ ions indirectly by becoming hydrated when dissolved in water (e.g. Al^{3+}, Fe^{3+}, Sn^{2+})	- Cation from strong bases and anion from weak acids (e.g. $NaCH_3COO$, KNO_2)
Salts (both acidic & basic) - Cation from weak bases and anion from weak acids (e.g. $NH_4CN \rightarrow NH_4^+ + CN^-$) *Check K_a & K_b to determine if the solution will be acidic/basic	**Salts (form neutral sol.)** - Cation from strong bases and anions from strong acids (e.g. NaCl, KCl, NaI, $NaNO_3$)

Metal Oxides + Water → Basic Solutions Non-metal Oxides + Water → Acidic Solutions

Titration is a technique used to find out the concentration of an unknown solution (analyte)
Equivalence Point: all moles of H_3O^+ have reacted with an equal number moles of OH^-
Buffers are solutions that can resist changes in pH when small amounts of acid/base are added
Requirements: weak acid + conjugate base / weak base + conjugate acid, and equal/high concentrations of both chemicals

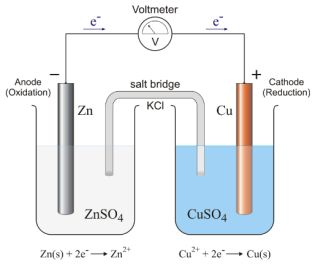
Strong-Strong Titrations: equivalence point at pH 7 (strong acid + strong base)
Weak-Strong Titrations: equivalence point above pH 7 (weak acid + strong base) or below pH 7 (strong acid + weak base)

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18 - Electrochemistry

Redox Reaction is a chemical reaction where electrons are transferred
Oxidation Numbers (ON) are used to track electron movement
*Oxidation = loss of electrons, Reduction = gain of electrons Oxidizing Agent = Reduced Substance, Reducing Agent = Oxidized Substance
*ON: Elemental State = 0, Oxygen = -2, Hydrogen = +1 (-1 with metals), Monoatomic Ions = Charge, Sum of ON = Charge of Compound

Galvanic (Voltaic) Cells is an electrochemical device that generates electrical energy from spontaneous redox reactions, electrons flow from anode (oxidation) to cathode (reduction)
Salt Bridge allows balancing of charges on both sides by moving cations to cathode and anion to anode
*Anode shrinks due to oxidation and Cathode grows due to reduction overtime



Standard Reduction Potentials (E^o) is the measure of a substance's ability to be reduced
Cell Potential (E^o_{cell}) measures the overall voltage of a complete electrochemical cell

$$E^o_{cell} = E^o_{cathode} + E^o_{anode}$$

*All values are compared to standard hydrogen electrode (SHE), which is assigned 0.00V
*↑ E^o = more likely to be reduced = strong oxidizing agent, ↓ E^o = more likely to be oxidized = strong reducing agent
*$E^o_{cell} > 0$ → spontaneous (reaction will proceed), $E^o_{cell} < 0$ → non-spontaneous (reaction will not proceed)